

std::quantity as a numeric type

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§ 1 Revision history

§ 1.1 Changes since [\[P2982R0\]](#)

- Fuel consumption example extended in [Converting between quantities of the same kind](#).
- [Dimension is not enough to describe a quantity](#) extended with [\[Measurement Data\]](#).
- [Lack of convertibility from fundamental types](#) added.
- [Terms and definitions](#) extended with the “Glossary” of [\[mp-units\]](#).
- Code samples in [Equivalence](#) reworked.
- Some small editorial fixes.

§ 2 Introduction

This document discusses the numeric and arithmetic aspects of the physical quantities and units library. This subject is broader than it could be initially imagined. Arithmetic operations are not only defined for user-facing types like `quantity` and `quantity_point`, but also for units and their magnitudes, dimensions, quantity specifications, and references. Each has its own requirements and constraints, and we will describe them in the following chapters.

Note: The code examples presented in this paper may not exactly reflect the final interface design that is going to be proposed in the follow-up papers. We are still doing some small fine-tuning to improve the library.

§ 3 Terms and definitions

This document consistently uses the official metrology vocabulary defined in the [\[ISO/IEC Guide 99\]](#) and [\[JCGM 200:2012\]](#). The most important of those and also [\[mp-units\]](#) implementation-specific terms can be found in [the “Glossary” chapter of the project’s documentation](#).

§ 4 Systems of quantities

The physical units libraries on the market typically only focus on modeling one or more systems of units. However, this is not the only system kind to model. Another, and maybe even more important is a system of quantities. The most important example here is the International System of Quantities (ISQ) defined by the series of [\[ISO/IEC 80000\]](#) documents.

§ 4.1 Dimension is not enough to describe a quantity

Most of the products on the market are aware of physical dimensions. However, a dimension is not enough to describe a quantity. This has been known for a long time now. The [\[Measurement Data\]](#) report from 1996 says explicitly, “Dimensional analysis does not adequately model the semantics of measurement data”.

A typical problem that most similar libraries struggle with is supporting quantities like work and torque as being independent, strong types. The problem here arises from the fact that both of them have exactly the same dimension L^2MT^{-2} , but a totally different physical meaning. As a result, it is possible to mathematically add or compare them in a quantity equation even though such an operation has no sense from the physical point of view.

A similar question that we could ask ourselves is what should be the result of:

```
auto res = 1 * Hz + 1 * Bq + 1 * Bd;
```

where:

- Hz (hertz) - a unit of frequency
- Bq (becquerel) - a unit of activity
- Bd (baud) - a unit of modulation rate

All of those quantities have the same dimension, namely T^{-1} , but it is probably not wise to allow adding, subtracting, or comparing them, as they describe vastly different physical properties.

It also is really tricky to separate handling of the quantities of dimension one. For example, angular measure expressed in radians and solid angular measure expressed in steradians should be independent. Any attempts to add or compare them should be detected at compile-time.

Last but not least, let's see the following implementation:

```
class Box {  
    quantity<square(si::metre)> base_;  
    quantity<si::metre> height_;  
public:  
    Box(quantity<si::metre> l, quantity<si::metre> w, quantity<si::metre> h)  
        : base_(l * w), height_(h) {}  
};
```

```
// ...  
};  
  
Box my_box(2 * m, 3 * m, 1 * m);
```

The above interface is far from being ideal. It does not provide any type-safety and enables potentially severe errors caused by accidental reordering of the constructor's arguments.

It turns out that the above issues can't be solved correctly without proper modeling of systems of quantities.

§ 4.2 Quantities of the same kind

The [ISO/IEC Guide 99] says:

- Quantities may be grouped together into categories of quantities that are **mutually comparable**
- Mutually comparable quantities are called **quantities of the same kind**
- Two or more quantities **cannot be added or subtracted unless they belong to the same category of mutually comparable quantities**
- Quantities of the **same kind** within a given system of quantities **have the same quantity dimension**
- Quantities of the **same dimension are not necessarily of the same kind**

[ISO/IEC 80000] also explicitly notes:

Measurement units of quantities of the same quantity dimension may be designated by the same name and symbol even when the quantities are not of the same kind. For example, joule per kelvin and J/K are respectively the name and symbol of both a measurement unit of heat capacity and a measurement unit of entropy, which are generally not considered to be quantities of the same kind. **However, in some cases special measurement unit names are restricted to be used with quantities of specific kind only.** For example, the measurement unit 'second to the power minus one' (1/s) is called hertz (Hz) when used for frequencies and becquerel (Bq) when used for activities of radionuclides. As another example, the joule (J) is used as a unit of energy, but never as a unit of moment of force, i.e. the newton metre (N · m).

Those provide answers to all the issues mentioned above. More than one quantity may be defined for the same dimension:

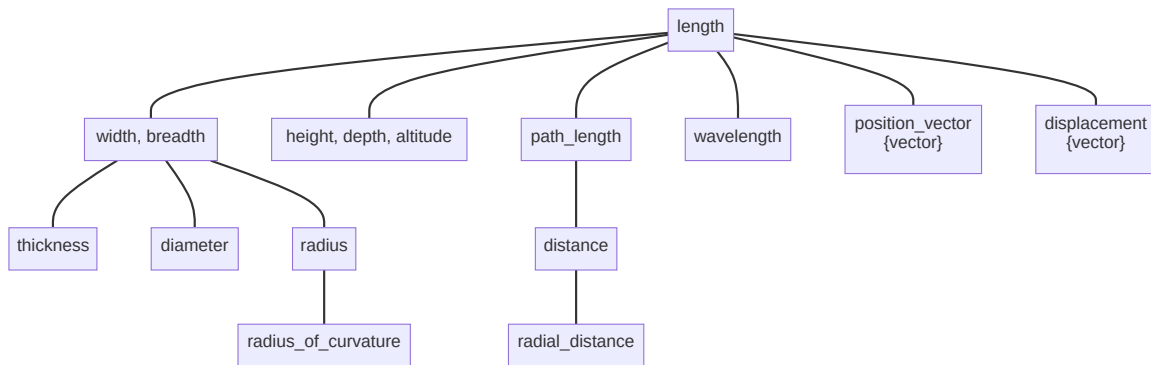
- quantities of different kinds (e.g. frequency, modulation rate, activity)
- quantities of the same kind (e.g. length, width, altitude, distance, radius, wavelength, position vector)

Two quantities can't be added, subtracted, or compared unless they belong to the same quantity kind.

§ 4.3 System of quantities is not only about kinds

The [ISO/IEC 80000] specifies hundreds of different quantities. Plenty of various kinds are provided, and often, each kind contains more than one quantity. It turns out that such quantities form a hierarchy of quantities of the same kind.

For example, here are all quantities of the kind length provided in [ISO/IEC 80000] (part 1):



Each of the above quantities expresses some kind of length, and each can be measured with meters, which is the unit defined by the [SI] for quantities of length. However, each has different properties, usage, and sometimes even a different character (position vector and displacement are vector quantities).

The below presents how such a hierarchy tree can be defined in the [mp-units] library:

```
inline constexpr struct dim_length : base_dimension<"L"> {} dim_length;

inline constexpr struct length : quantity_spec<dim_length> {} length;
inline constexpr struct width : quantity_spec<length> {} width;
inline constexpr auto breadth = width;
inline constexpr struct height : quantity_spec<length> {} height;
inline constexpr auto depth = height;
inline constexpr auto altitude = height;
inline constexpr struct thickness : quantity_spec<width> {} thickness;
inline constexpr struct diameter : quantity_spec<width> {} diameter;
inline constexpr struct radius : quantity_spec<width> {} radius;
inline constexpr struct radius_of_curvature : quantity_spec<radius> {}
    radius_of_curvature;
inline constexpr struct path_length : quantity_spec<length> {}
    path_length;
inline constexpr auto arc_length = path_length;
inline constexpr struct distance : quantity_spec<path_length> {} distance;
inline constexpr struct radial_distance : quantity_spec<distance> {}
    radial_distance;
inline constexpr struct wavelength : quantity_spec<length> {} wavelength;
```

```
inline constexpr struct position_vector : quantity_spec<length,
quantity_character::vector> {} position_vector;
inline constexpr struct displacement : quantity_spec<length,
quantity_character::vector> {} displacement;
```

In the above code:

- length takes the base dimension to indicate that we are creating a base quantity that will serve as a root for a tree of quantities of the same kind,
- width and following quantities are branches and leaves of this tree with the parent always provided as the argument to quantity_spec class template,
- breadth is an alias name for the same quantity as width.

Please note that some quantities may be specified by [ISO/IEC 80000] as vector or tensor quantities (e.g. displacement).

§ 4.4 Converting between quantities of the same kind

Quantity conversion rules can be defined based on the same hierarchy of quantities of kind length.

1. Implicit conversions

- Every width is a length.
- Every radius is a width.

```
static_assert(implicitly_convertible(isq::width, isq::length));
static_assert(implicitly_convertible(isq::radius, isq::length));
static_assert(implicitly_convertible(isq::radius, isq::width));
```

In the [mp-units] library, implicit conversions are allowed on copy-initialization:

```
void foo(quantity<isq::length<m>> q);

quantity<isq::width<m>> q1 = 42 * m;
quantity<isq::length<m>> q2 = q1; // implicit quantity conversion
foo(q1);                          // implicit quantity conversion
```

2. Explicit conversions

- Not every length is a width.
- Not every width is a radius.

```
static_assert(!implicitly_convertible(isq::length, isq::width));
static_assert(!implicitly_convertible(isq::length, isq::radius));
static_assert(!implicitly_convertible(isq::width, isq::radius));
static_assert(implicitly_convertible(isq::length, isq::width));
static_assert(implicitly_convertible(isq::length, isq::radius));
static_assert(implicitly_convertible(isq::width, isq::radius));
```


In the [mp-units] library, explicit conversions are forced by passing the quantity to a call operator of a quantity_spec type:

```
quantity<isq::length<m>> q1 = 42 * m;  
quantity<isq::height<m>> q2 = isq::height(q1); // explicit quantity  
conversion
```

3. Explicit casts

- height is never a width, and vice versa.
- Both height and width are quantities of kind length.

```
static_assert(!implicitly_convertible(isq::height, isq::width));  
static_assert(!explicitly_convertible(isq::height, isq::width));  
static_assert(!castable(isq::height, isq::width));
```

In the [mp-units] library, explicit casts are forced with a dedicated quantity_cast function:

```
quantity<isq::width<m>> q1 = 42 * m;  
quantity<isq::height<m>> q2 = quantity_cast<isq::height>(q1); //  
explicit quantity cast
```

4. No conversion

- time has nothing in common with length.

```
static_assert(!implicitly_convertible(isq::time, isq::length));  
static_assert(!explicitly_convertible(isq::time, isq::length));  
static_assert(!castable(isq::time, isq::length));
```

In the [mp-units] library, even the explicit casts will not force such a conversion:

```
void foo(quantity<isq::length[m]>);  
  
foo(quantity_cast<isq::length>(42 * s)); // Compile-time error
```

With the above rules, one can write the following short application to calculate a fuel consumption:

```
inline constexpr struct fuel_volume : quantity_spec<isq::volume> {}  
fuel_volume;  
inline constexpr struct fuel_consumption : quantity_spec<fuel_volume /  
isq::distance> {} fuel_consumption;  
  
const quantity fuel = fuel_volume(40. * l);  
const quantity distance = isq::distance(550. * km);  
const quantity<fuel_consumption[l / (mag<100> * km)]> q = fuel / distance;  
std::cout << "Fuel consumption: " << q << "\n";
```

The above code prints:

Fuel consumption: 7.27273×10^{-2} l/km

Please note that, despite the dimensions of `fuel_consumption` and `isq::area` being the same (L^2), the constructor of a quantity `q` below will fail to compile when we pass an argument being the quantity of area:

```
static_assert(fuel_consumption.dimension == isq::area.dimension);

const quantity<isq::area[m2]> football_field = isq::length(105 * m) *
    isq::width(68 * m);
const quantity<fuel_consumption[l / (mag<100> * km)]> q2 = football_field;
    // Compile-time error
const quantity q3 = q + football_field;
    // Compile-time error
if (q == football_field) {
    // Compile-time error
    // ...
}
```

§ 4.4.1 Comparing, adding, and subtracting quantities of the same kind

[ISO/IEC Guide 99] explicitly states that width and height are quantities of the same kind and as such they

- are mutually comparable, and
- can be added and subtracted.

If we take the above for granted, the only reasonable result of `1 * width + 1 * height` is `2 * length`, where the result of `length` is known as a common quantity type. A result of such an equation is always the first common node in a hierarchy tree of the same kind. For example:

```
static_assert(common_quantity_spec(isq::width, isq::height) ==
    isq::length);
static_assert(common_quantity_spec(isq::thickness, isq::radius) ==
    isq::width);
static_assert(common_quantity_spec(isq::distance, isq::path_length) ==
    isq::path_length);

quantity q = isq::thickness(1 * m) + isq::radius(1 * m);
static_assert(q.quantity_spec == isq::width);
```

One could argue that allowing to add or compare quantities of height and width might be a safety issue, but we need to be consistent with the requirements of [ISO/IEC 80000]. Moreover, from our experience, disallowing such operations and requiring an explicit cast to a common quantity in every single place makes the code so cluttered with casts that it nearly renders the library unusable.

Fortunately, the above-mentioned conversion rules make the code safe by construction anyway. Let's analyze the following example:

```
inline constexpr struct horizontal_length : quantity_spec<isq::length> {}
    horizontal_length;

namespace christmas {
```

```

struct gift {
    quantity<horizontal_length[m]> length;
    quantity<isq::width[m]> width;
    quantity<isq::height[m]> height;
};

std::array<quantity<isq::length[m]>, 2> gift_wrapping_paper_size(const
    gift& g)
{
    const auto dim1 = 2 * g.width + 2 * g.height + 0.5 * g.width;
    const auto dim2 = g.length + 2 * 0.75 * g.height;
    return { dim1, dim2 };
}

} // namespace christmas

int main()
{
    const christmas::gift lego = { horizontal_length(40 * cm), isq::width(30
        * cm), isq::height(15 * cm) };
    auto paper = christmas::gift_wrapping_paper_size(lego);

    std::cout << "Paper needed to pack a lego box:\n";
    std::cout << "- " << paper[0] << " X " << paper[1] << "\n"; // - 1.05 m
        X 0.625 m
    std::cout << "- area = " << paper[0] * paper[1] << "\n"; // - area =
        0.65625 m2
}

```

In the beginning, we introduce a custom quantity `horizontal_length` of a kind `length`, which then, together with `isq::width` and `isq::height`, are used to define the dimensions of a Christmas gift. Next, we provide a function that calculates the dimensions of a gift wrapping paper with some wraparound. The result of both those expressions is a quantity of `isq::length`, as this is the closest common quantity for the arguments used in this quantity equation.

Regarding safety, it is important to mention here, that thanks to the conversion rules provided above, it would be impossible to accidentally do the following:

```

void foo(quantity<horizontal_length[m]> q);

quantity<isq::width[m]> q1 = dim1; // Compile-time error
quantity<isq::height[m]> q2{dim1}; // Compile-time error
foo(dim1); // Compile-time error

```

The reason of compilation errors above is the fact that `isq::length` is not implicitly convertible to the quantities defined based on it. To make the above code compile, an explicit conversion of a quantity type is needed:

```

void foo(quantity<horizontal_length[m]> q);

quantity<isq::width[m]> q1 = isq::width(dim1);

```

```
quantity<isq::height[m]> q2{isq::height(dim1)};
foo(horizontal_length(dim1));
```

To summarize, rules for addition, subtraction, and comparison of quantities improve the library usability, while the conversion rules enhance the safety of the library compared to the libraries that do not model quantity kinds.

§ 4.5 Hierarchies of derived quantities

The same rules propagate to derived quantities. For example, we can define strongly typed horizontal length and area:

```
inline constexpr struct horizontal_length : quantity_spec<isq::length> {}
    horizontal_length;
inline constexpr struct horizontal_area : quantity_spec<isq::area,
    horizontal_length * isq::width> {} horizontal_area;
```

The first definition says that a horizontal length is a more specialized quantity than length and belongs to the same quantity kind. The second line defines a horizontal area, which is a more specialized quantity than area, so it has a more constrained recipe as well. Thanks to that:

```
static_assert(implicitly_convertible(horizontal_length, isq::length));
static_assert(!implicitly_convertible(isq::length, horizontal_length));
static_assert(explicitly_convertible(isq::length, horizontal_length));

static_assert(implicitly_convertible(horizontal_area, isq::area));
static_assert(!implicitly_convertible(isq::area, horizontal_area));
static_assert(explicitly_convertible(isq::area, horizontal_area));

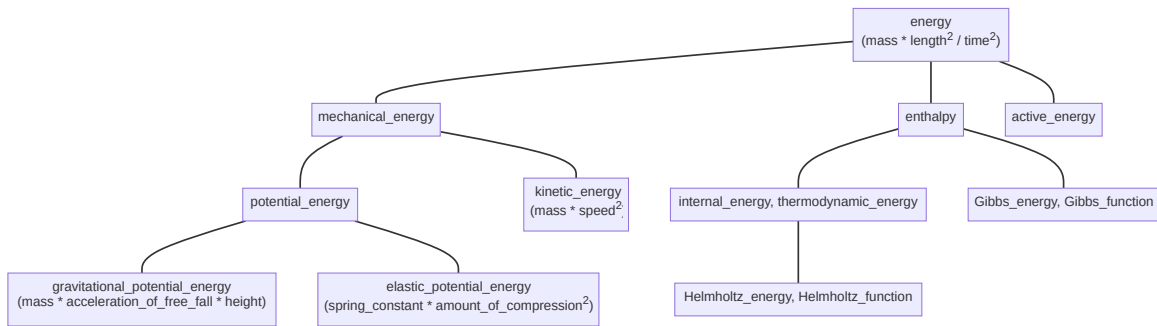
static_assert(implicitly_convertible(isq::length * isq::length,
    isq::area));
static_assert(!implicitly_convertible(isq::length * isq::length,
    horizontal_area));
static_assert(explicitly_convertible(isq::length * isq::length,
    horizontal_area));

static_assert(implicitly_convertible(horizontal_length * isq::width,
    isq::area));
static_assert(implicitly_convertible(horizontal_length * isq::width,
    horizontal_area));
```

Unfortunately, derived quantity equations often do not automatically form a hierarchy tree. This is why sometimes it is not obvious what such a tree should look like. Also, the [\[ISO/IEC Guide 99\]](#) explicitly states:

The division of ‘quantity’ according to ‘kind of quantity’ is, to some extent, arbitrary.

The below presents some arbitrary hierarchy of derived quantities of kind energy:



Notice, that even though all of those quantities have the same dimension and can be expressed in the same units, they have different quantity equations used to create them implicitly:

- energy is the most generic one and thus can be created from base quantities of mass, length, and time. As those are also the roots of quantities of their kinds and all other quantities are implicitly convertible to them, it means that an energy can be implicitly constructed from any quantity having proper powers of mass, length, and time.

```
static_assert(implicitly_convertible(isq::mass * pow<2>(isq::length) /
    pow<2>(isq::time), isq::energy));
static_assert(implicitly_convertible(isq::mass * pow<2>(isq::height) /
    pow<2>(isq::time), isq::energy));
```

- mechanical_energy is a more “specialized” quantity than energy (not every energy is a mechanical_energy). It is why an explicit cast is needed to convert from either energy or the results of its quantity equation.

```
static_assert(!implicitly_convertible(isq::energy,
    isq::mechanical_energy));
static_assert(explicitly_convertible(isq::energy,
    isq::mechanical_energy));
static_assert(!implicitly_convertible(isq::mass * pow<2>(isq::length)
    / pow<2>(isq::time), isq::mechanical_energy));
static_assert(explicitly_convertible(isq::mass * pow<2>(isq::length) /
    pow<2>(isq::time), isq::mechanical_energy));
```

- gravitational_potential_energy is not only even more specialized one but additionally, it is special in a way that it provides its own “constrained” quantity equation. Maybe not every $\text{mass} * \text{pow}<2>(\text{length}) / \text{pow}<2>(\text{time})$ is a gravitational_potential_energy, but every $\text{mass} * \text{acceleration_of_free_fall} * \text{height}$ is.

```
static_assert(!implicitly_convertible(isq::energy,
    gravitational_potential_energy));
static_assert(explicitly_convertible(isq::energy,
    gravitational_potential_energy));
static_assert(!implicitly_convertible(isq::mass * pow<2>(isq::length)
    / pow<2>(isq::time), gravitational_potential_energy));
static_assert(explicitly_convertible(isq::mass * pow<2>(isq::length) /
    pow<2>(isq::time), gravitational_potential_energy));
static_assert(implicitly_convertible(isq::mass * acceleration_of_free_fall * isq::height,
    gravitational_potential_energy));
```

§ 4.6 Modeling a quantity kind

In the physical units library, we also need an abstraction describing an entire family of quantities of the same kind. Such quantities have not only the same dimension but also can be expressed in the same units.

To annotate a quantity to represent its kind (and not just a hierarchy tree’s root quantity), in [mp-units] we introduced a `kind_of<>` specifier. For example, to express any quantity of length, we need to specify `kind_of<isq::length>`. That entity behaves as any quantity of its kind. This means that it is implicitly convertible to any quantity in a tree:

```
static_assert(!implicitly_convertible(isq::length, isq::height));
static_assert(implicitly_convertible(kind_of<isq::length>, isq::height));
```

Additionally, the result of operations on quantity kinds is also a quantity kind:

```
static_assert(same_type<kind_of<isq::length> / kind_of<isq::time>,
              kind_of<isq::length / isq::time>>);
```

However, if at least one equation’s operand is not a quantity kind, the result becomes a “strong” quantity where all the kinds are converted to the hierarchy tree’s root quantities:

```
static_assert(!same_type<kind_of<isq::length> / isq::time,
              kind_of<isq::length / isq::time>>);
static_assert(same_type<kind_of<isq::length> / isq::time, isq::length /
              isq::time>);
```

Please note that only a root quantity from the hierarchy tree or the one marked with `is_kind` specifier in the `quantity_spec` definition can be put as a template parameter to the `kind_of` specifier. For example, `kind_of<isq::width>` will fail to compile.

§ 5 Systems of units

Modeling a system of units is the most important feature and a selling point of every physical units library. Thanks to that, the library can protect users from performing invalid operations on quantities and provide automated conversion factors between various compatible units.

Probably all the libraries in the wild model the [SI], and many of them provide support for additional units belonging to various other systems (e.g. imperial).

§ 5.1 Systems of units are based on systems of quantities

Systems of quantities specify a set of quantities and equations relating to those quantities. Those equations do not take any unit or a numerical representation into account at all. In order to create a quantity, we need to add those missing pieces of information. This is where a system of units kicks in.

The [SI] is explicitly stated to be based on the ISQ. Among others, it defines seven base units, one for each base quantity. In [mp-units], this is expressed by associating a quantity kind to a unit being defined:

```
inline constexpr struct metre : named_unit<"m", kind_of<isq::length>> {}  
    metre;
```

The `kind_of<isq::length>` above states explicitly that this unit has an associated quantity kind. In other words, `si::metre` (and scaled units based on it) can be used to express the amount of any quantity of kind length.

Associated units are so useful and common in the [mp-units] library that they got their own concepts `AssociatedUnit<T>` to improve the interfaces.

Please note that for some systems of units (e.g., natural units), a unit may not have an associated quantity type. For example, if we define the speed of light constant as $c = 1$, we can define a system where both length and time will be measured in seconds, and speed will be a quantity measured with the unit one. In such case, the definition will look as follows:

```
inline constexpr struct second : named_unit<"s"> {} second;
```

§ 5.2 Units compose

One of the strongest points of the [SI] system is that its units compose. This allows providing thousands of different units for hundreds of various quantities with a really small set of predefined units and prefixes. For example, one can write:

```
quantity<si::metre / si::second> q;
```

to express a quantity of speed. The resulting quantity type is implicitly inferred from the unit equation by repeating exactly the same operations on the associated quantity kinds.

Also, as units are regular values, we can easily provide a helper ad-hoc unit with:

```
constexpr auto mps = si::metre / si::second;  
quantity<mps> q;
```

§ 5.3 Many shades of the same unit

The [SI] provides the names for 22 common coherent units of 22 derived quantities.

Each such named derived unit is a result of a specific predefined unit equation. For example, a unit of power quantity is defined as:

```
inline constexpr struct watt : named_unit<"W", joule / second> {} watt;
```

However, a power quantity can be expressed in other units as well. For example, the following:

```
auto q1 = 42 * W;
std::cout << q1 << "\n";
std::cout << q1.in(J / s) << "\n";
std::cout << q1.in(N * m / s) << "\n";
std::cout << q1.in(kg * m2 / s3) << "\n";
```

prints:

```
42 W
42 J/s
42 N m/s
42 kg m2/s3
```

All of the above quantities are equivalent and mean exactly the same.

§ 5.4 Constraining a derived unit to work only with a specific derived quantity

Some derived units are valid only for specific derived quantities. For example, [SI] specifies both hertz and becquerel derived units with the same unit equation 1/s. However, it also explicitly states:

The hertz shall only be used for periodic phenomena and the becquerel shall only be used for stochastic processes in activity referred to a radionuclide.

This is why it is important for the library to allow constraining such units to be used only with a specific quantity kind:

```
inline constexpr struct hertz : named_unit<"Hz", one / second,
    kind_of<isq::frequency>> {} hertz;
inline constexpr struct becquerel : named_unit<"Bq", one / second,
    kind_of<isq::activity>> {} becquerel;
```

With the above, hertz can only be used for frequencies, while becquerel should only be used for quantities of activity. This means that the following equation will not compile, improving the type-safety of the library:

```
auto q = 1 * Hz + 1 * Bq; // Fails to compile
```

§ 5.5 Prefixed units

Besides named units, the SI specifies also 24 prefixes (all being a power of 10) that can be prepended to all named units to obtain various scaled versions of them.

Implementation of std::ratio provided by all major compilers is able to express only 16 of them. This is why, in the [mp-units], we had to find an alternative way to represent a unit's magnitude in a more flexible way.

Each prefix is implemented as:


```
template<PrefixableUnit auto U> struct quecto_ : prefixed_unit<"q",
    mag_power<10, -30>, U> {};
template<PrefixableUnit auto U> inline constexpr quecto_<U> quecto;
```

and then a unit can be prefixed in the following way:

```
inline constexpr auto qm = quecto<metre>;
```

The usage of `mag_power` not only enables providing support for SI prefixes, but it can also efficiently represent any rational magnitude. For example, [ISO/IEC 80000] (part 13) prefixes used in the IT industry can be implemented as:

```
template<PrefixableUnit auto U> struct yobi_ : prefixed_unit<"Yi",
    mag_power<2, 80>, U> {};
template<PrefixableUnit auto U> inline constexpr yobi_<U> yobi;
```

Please note that we need two lines with two definitions in the case of a class template and an associated variable template. The C and C++ standards permit providing the same identifier for a class and its variable and appending the variable name after the class definition. None of this is allowed for class templates.

§ 5.6 Scaled units

In the [SI], all units are either base or derived units or prefixed versions of those. However, those are not the only options possible.

For example, there is a list of off-system units accepted for use with [SI]. All of those are scaled versions of the [SI] units with ratios that can't be explicitly expressed with predefined SI prefixes. Those include units like minute, hour, or electronvolt:

```
inline constexpr struct minute : named_unit<"min", mag<60> * si::second>
    {} minute;
inline constexpr struct hour : named_unit<"h", mag<60> * minute> {} hour;
inline constexpr struct electronvolt : named_unit<"eV",
    mag<ratio{1'602'176'634, 1'000'000'000}> * mag_power<10, -19> *
    si::joule> {} electronvolt;
```

Also, units of other systems of units are often defined in terms of scaled versions of other (often SI) units. For example, the international yard is defined as:

```
inline constexpr struct yard : named_unit<"yd", mag<ratio{9'144, 10'000}>
    * si::metre> {} yard;
```

and then a foot can be defined as:

```
inline constexpr struct foot : named_unit<"ft", mag<ratio{1, 3}> * yard>
    {} foot;
```

For some units, a magnitude might also be irrational. The best example here is a degree which is defined using a floating-point magnitude having a factor of the number π (Pi):

```
inline constexpr struct mag_pi : magnitude<std::numbers::pi_v<long
    double>> {} mag_pi;
```

```
inline constexpr struct degree : named_unit<basic_symbol_text{"°", "deg"},
    mag_pi / mag<180> * si::radian> {} degree;
```

§ 5.7 Unit symbols

In the [mp-units] library, units are available via their full names or through their short symbols. To use a long version it is enough to type:

```
#include <mp-units/systems/si/si.h>

using namespace mp_units;

quantity q = 42 * si::metre;
```

The same can be obtained using an optional unit symbol:

```
#include <mp-units/systems/si/si.h>

using namespace mp_units;
using namespace mp_units::si::unit_symbols;

quantity q = 42 * m;
```

Unit symbols introduce a lot of short identifiers into the current namespace, and that is why they are opt-in. A user has to explicitly “import” them from a dedicated `unit_symbols` namespace.

§ 6 Operations on units, dimensions, and quantity types

Modern C++ physical quantities and units library should expose compile-time constants for units, dimensions, and quantity types. Each of such constants should be of a different type. Said otherwise, every unit, dimension, and quantity type has a unique type and a compile-time instance. This allows us to do regular algebra on such identifiers and get proper types as results of such operations.

The operations exposed by such a library should include at least:

- multiplication (e.g. `newton * metre`),
- division (e.g. `metre / second`),
- power (e.g. `pow<2>(metre)` or `pow<1, 2>(metre * metre)`).

To improve the usability of the library, we also recommend adding:

- square root (e.g. `sqrt(metre * metre)` as equivalent to `pow<1, 2>(metre * metre)`),
- cubic root (e.g. `cbrt(metre * metre * metre)` as equivalent to `pow<1, 3>(metre * metre * metre)`),

— inversion (e.g. `inverse(second)` as equivalent to `one / second`).

Additionally, for units only, to improve the readability of the code, it makes sense to expose the following:

— square power (e.g. `square(metre)` is equivalent to `pow<2>(metre)`),

— cubic power (e.g. `cubic(metre)` is equivalent to `pow<3>(metre)`).

The above two functions could also be considered for dimensions and quantity types. However, `cubic(length)` does not seem to make much sense, and probably `pow<3>(length)` should be preferred instead.

§ 6.1 Expression templates

Modern C++ physical quantities and units libraries use opaque types to improve the user experience while analyzing compile-time errors or inspecting types in a debugger. This is a huge usability improvement over the older libraries that use aliases to refer to long instantiations of class templates.

§ 6.1.1 Derived entities

Having such strong types for entities is not enough. While doing arithmetics on them, we get derived entities, and they also should be easy to understand and correlate with the code written by the user. This is where expression templates come into play.

The library should use the same unified approach to represent the results of arithmetics on all kinds of entities. It is worth mentioning that a generic purpose expression templates library is not a good solution for a physical quantities and units library.

Let's assume that we want to represent the results of the following two unit equations:

— `metre / second * second`

— `metre * metre / metre`

Both of them should result in a type equivalent to `metre`. A general-purpose library will probably result with the types similar to the below:

— `mul<div<metre, second>, second>`

— `div<mul<metre, metre>, metre>`

Comparing such types for equivalence would not only be very expensive at compile-time but would also be really confusing to the users observing them in the compilation logs. This is why we need a dedicated solution here.

In a physical quantities and units library, we need expression templates to express the results of

— dimension equations,

- quantity type equations,
- unit equations, and
- unit magnitude equations.

If the above equation results in a derived entity, we must create a type that clearly describes what we are dealing with. We need to pack a simplified expression template into some container for that. There are various possibilities here. The table below presents the types generated from unit expressions by two leading products on the market in this subject:

Unit	[mp-units]	[Au]
N·m	derived_unit<metre, newton>	UnitProduct<Meters, Newtons>
1/s	derived_unit<one, per<second>>	Pow<Seconds, -1>
km/h	derived_unit<kilo_<metre>, per<hour>>	UnitProduct<Kilo<Meters>, Pow<Hours, -1>>
kg·m ² /(s ³ ·K)	derived_unit<kilogram, pow<metre, 2>, per<kelvin, power<second, 3>>>	UnitProduct<Pow<Meters, 2>, Kilo<Grams>, Pow<Seconds, -3>, Pow<Kelvins, -1>>
m ² /m	metre	Meters
km/m	derived_unit<kilo_<metre>, per<metre>>	UnitProduct<Pow<Meters, -1>, Kilo<Meters>>
m/m	one	UnitProduct<>

It is a matter of taste which solution is better. While discussing the pros and cons here, we should remember that our users often do not have a scientific background. This is why the [mp-units] library decided to use syntax that is as similar to the correct English language as possible. It consistently uses the `derived_` prefix for types representing derived units, dimensions, and quantity specifications. Those are instantiated first with the contents of the numerator followed by the entities of the denominator (if present) enclosed in the `per<...>` expression template.

§ 6.1.2 Identities

The arithmetics on units, dimensions, and quantity types require a special identity value. Such value can be returned as a result of the division of the same entities, or using it should not modify the expression template on multiplication.

The [mp-units] library chose the following names here:

- `one` in the domain of units,
- `dimension_one` in the domain of dimensions,

— dimensionless in the domain of quantity types.

The above names were selected based on the following quote from the [ISO/IEC 80000]:

A quantity whose dimensional exponents are all equal to zero has the dimensional product denoted $A^0B^0C^0\dots = 1$, where the symbol 1 denotes the corresponding dimension. There is no agreement on how to refer to such quantities. They have been called **dimensionless** quantities (although this term should now be avoided), quantities with **dimension one**, quantities with dimension number, or quantities with the **unit one**. Such quantities are dimensionally simply numbers. To avoid confusion, it is helpful to use explicit units with these quantities where possible, e.g., m/m, nmol/mol, rad, as specified in the SI Brochure.

§ 6.1.3 Supported operations and their results

The table below presents all the operations that can be done on units, dimensions, and quantity types in a physical quantities and units library and corresponding expression templates chosen by the [mp-units] project as their results:

Operation	Resulting template expression arguments
$A * B$	A, B
$B * A$	A, B
$A * A$	power<A, 2>
$\{\text{identity}\} * A$	A
$A * \{\text{identity}\}$	A
A / B	A, per
A / A	{identity}
$A / \{\text{identity}\}$	A
$\{\text{identity}\} / A$	{identity}, per<A>
$\text{pow}<2>(A)$	power<A, 2>
$\text{pow}<2>(\{\text{identity}\})$	{identity}
$\text{sqrt}(A)$ or $\text{pow}<1, 2>(A)$	power<A, 1, 2>
$\text{sqrt}(\{\text{identity}\})$ or $\text{pow}<1, 2>(\{\text{identity}\})$	{identity}

§ 6.1.4 Simplifying the resulting expression templates

To limit the length and improve the readability of generated types, there are many rules to simplify the resulting expression template.

1. Ordering

The resulting comma-separated arguments of multiplication are always sorted according to a specific predicate. This is why:

```
static_assert(A * B == B * A);  
static_assert(std::is_same_v<decltype(A * B), decltype(B * A)>);
```

This is probably the most important of all the steps, as it allows comparing types and enables the rest of the simplification rules.

Units and dimensions have unique symbols, but ordering quantity types might not be that trivial. Although the ISQ defined in [ISO/IEC 80000] provides symbols for each quantity, there is little use for them in the C++ code. This is caused by the fact that such symbols use a lot of characters that are not available with the Unicode encoding. Most of the limitations correspond to Unicode providing only a minimal set of characters available as subscripts, which are often used to differentiate various quantities of the same kind. For example, it is impossible to encode the symbols of the following quantities:

- c_{sat} - specific heat capacity at saturated vapour pressure,
- μ_{JT} - Joule-Thomson coefficient,
- $w_{\text{H}_2\text{O}}$ - mass fraction of water,
- $\sigma_{\Omega, E}$ - direction and energy distribution of cross section,
- $d_{1/2}$ - half-value thickness,
- $\Phi_{e, \lambda}$ - spectral radiant flux.

This is why the [mp-units] library chose to use type name identifiers in such cases.

2. Aggregation

In case two of the same type identifiers are found next to each other on the argument list, they will be aggregated in one entry:

Before	After
A, A	power<A, 2>
A, power<A, 2>	power<A, 3>
power<A, 1, 2>, power<A, 2>	power<A, 5, 2>
power<A, 1, 2>, power<A, 1, 2>	A

3. Simplification

In case two of the same type identifiers are found in the numerator and denominator argument lists, they are being simplified into one entry:

Before	After
A, per<A>	{identity}
power<A, 2>, per<A>	A
power<A, 3>, per<A>	power<A, 2>
A, per<power<A, 2>>	{identity}, per<A>

It is important to notice here that only the elements with exactly the same type are being simplified. This means that, for example, m/m results in one, but km/m will not be simplified. The resulting derived unit will preserve both symbols and their relative magnitude. This allows us to properly print symbols of some units or constants that require such behavior. For example, the Hubble constant is expressed in $\text{km}\cdot\text{s}^{-1}\cdot\text{Mpc}^{-1}$, where both km and Mpc are units of length.

4. Repacking

In case an expression uses two results of some other operations, the components of its arguments are repacked into one resulting type and simplified there.

For example, assuming:

```
constexpr auto X = A / B;
```

then:

Operation	Resulting template expression arguments
X * B	A
X * A	power<A, 2>, per
X * X	power<A, 2>, per<power<B, 2>>
X / X	{identity}
X / A	{identity}, per
X / B	A, per<power<B, 2>>

Please note that for as long as for the ordering step in some cases, we use user-provided symbols, the aggregation, and the next steps do not benefit from those. They always use type identifiers to determine whether the operation should be performed.

Unit symbols are not guaranteed to be unique in the project. For example, someone may use "s" as a symbol for a count of samples, which, when used in a unit expression with seconds, would cause fatal consequences (e.g. sample * second would yield s², or sample / second would result in one).

Some units would provide worse text output if the ordering step used type identifiers rather than unit symbols. For example, si::metre * si::second * cgs::second would result in s m s, or newton * metre would result in m N, which is not how we typically spell this unit. However, for the sake of consistency, we may also consider changing the algorithm used for ordering to be based on type identifiers.

§ 6.1.5 Example

Thanks to all of the steps described above, a user may write the code like this one:

```
using namespace mp_units::si::unit_symbols;
quantity speed = isq::speed(60. * km / h);
quantity duration = 8 * s;
quantity acceleration1 = speed / duration;
quantity acceleration2 = isq::acceleration(acceleration1.in(m / s2));
std::cout << "acceleration: " << acceleration1 << " (" << acceleration2 <<
    ")\n";
```

the text output provides:

acceleration: 7.5 km h⁻¹ s⁻¹ (2.08333 m/s²)

The above program will produce the following types for acceleration quantities (after stripping the `mp_units` namespace for brevity):

— acceleration1

```
quantity<reference<derived_quantity_spec<isq::speed,    per<isq::time>>
    {},
                                     derived_unit<si::kilo_<si::metre{}>,
    per<non_si::hour, si::second>>{}>{},
    double>
```

— acceleration2

```
quantity<reference<isq::acceleration,
    derived_unit<si::metre, per<power<si::second, 2>>>
    {}>{},
    double>>
```

§ 6.2 Scaled units

Another very common operation is to multiply an existing unit by a factor, creating a new, scaled unit. For example, the unit *foot* can be multiplied by 3, producing the unit *yard*.

The process also works in reverse; the ratio between any two units of the same dimension is a well-defined number. For example, the ratio between one foot and one inch is 12.

§ 6.2.1 Unit magnitudes

In principle, this scaling factor can be any positive real number. In `[mp-units]` and `[Au]`, we have used the term “magnitude” to refer to this scaling factor. (This should not be confused with other uses of the term, such as the logarithmic “magnitude” unit commonly used in astronomy).

In the library implementation, each unit is associated with a magnitude. However, for most units, the magnitude is a fully encapsulated implementation detail, not a user-facing value.

This is because the notion of “the” magnitude of a unit is not generally meaningful: it has no physically observable consequence. What *is* meaningful is the *ratio* of magnitudes between two units of the same quantity kind. We could associate the *foot*, say, with any magnitude that we like — but once we make that choice, we must assign to the *yard*, and to the *inch*. Separately and independently, we can assign any magnitude to the second, because it’s an independent dimension — but once we make that choice, it fixes the magnitude for derived units, and we must assign, say, to the *mile per hour*.

§ 6.2.2 Requirements and representation

A magnitude is a positive real number. The best way to *represent* it depends on how we will *use* it. To derive our requirements, note that magnitudes must support every operation which units do.

Units are closed under *products* and *rational powers*. Therefore, our magnitude representation must support these operations natively and robustly; this is the most basic requirement. We must also support certain *irrational* “ratios”, such as the factor of — between *degrees* and *radians*.

The usual approach, `std::ratio`, fails to satisfy these requirements in multiple ways.

- It is not closed under rational powers (rather infamously, in the case of $2^{1/2}$).
- It cannot represent irrational factors such as π .
- It is too vulnerable to overflow when raised to powers.

One alternative is the *vector space magnitude* representation. Here, we represent each magnitude as a *product of powers of “basis” numbers*. This is the same representation we use for dimensions, so it will naturally support all the same operations — as long as we can find a suitable basis.

Each magnitude must have a unique representation. This requirement constrains our choice of “basis” vectors; they must not be able to represent any magnitude in more than one way. Prime numbers have this property. Take any arbitrarily large (but finite) collection of primes, raise each prime to some chosen exponent, and compute the product: the result can’t be expressed by any other collection of exponents.

Already, this lets us represent every positive real number which `std::ratio` can represent, by breaking the numerator and denominator into their prime factorizations. But we can go further, and handle irrational factors such as π by introducing them as new basis vectors. π cannot be represented by the product of powers of *any* finite collection of primes, which means that it is “linearly independent” in the sense of our vector space representation.

On the C++ implementation side, we use variadic templates to define our magnitude. Each element is a basis number raised to some rational power (which may be omitted or abbreviated as appropriate).

Here are some examples, using Astronomical Units (au), meters (m), degrees (deg), and radians (rad).

Unit ratio	std::ratio representation	vector space representation
—	std::ratio<149'597'870'700>	magnitude<power_v<2, 2>(), 3, power_v<5, 2>(), 73, 877, 7789>
—	Unrepresentable (overflow)	magnitude<power_v<2, 4>(), power_v<3, 2>(), power_v<5, 4>(), power_v<73, 2>(), power_v<877, 2> (), power_v<7789, 2>(>
—	Unrepresentable	magnitude<2, power_v<3, 1, 2>(), 5, power_v<73, 1, 2>(), power_v<877, 1, 2>(), power_v<7789, 1, 2>(>
—	Unrepresentable	magnitude<power_v<2, 2>(), power_v<3, 2>(), power_v<3.14159265358979323851e+0l, -1>(), 5>

The variadic magnitude types have one disadvantage: they are more verbose. We may be able to hide them via opaque types with nicer names, using a similar strategy as we have for units. In any case, their major advantage is that they fulfill the requirements stated above — indeed, they are the only solution we have seen which does.

§ **6.2.3 Construction helpers**

We do not want end users to specify the variadic magnitude implementation manually. That would be too labor intensive and error prone. Instead, we provide *construction helpers* for end users.

mag<N> creates the vector space representation of the integer N. This is a simple value, so we can multiply and divide it with other magnitudes.

For example, mag<180> / mag<240> produces magnitude<power_v<2, -2>(), 3>, which is $\frac{1}{2}$. Notice how the result is always automatically represented in lowest terms, because a fraction *not* in lowest terms cannot be represented as a product of powers of primes.

§ **6.2.4 Common units**

Some operations, such as addition or inequality comparison, are “common unit” operations. In order to execute them on quantities of the same dimension, we must first convert them to their *common unit*.

For units which are *commensurable* — that is, units whose ratio is a rational number — the “common unit” is the largest unit that evenly divides both. For example, the common unit of the *foot* and *inch* is the inch.

This also includes instances where neither unit is an integer multiple of the other. For example, the common unit of the meter and the yard does not have a name, but is equivalent to 800 micrometers. If we call this unit u , then there are 1250 u per meter, and 1143 u per yard.

The practical benefit of this definition is that this unit conversion will simply multiply each participating quantity by an exact integer. If we use integer types to begin with, we can continue to use them without losing precision.

Unfortunately, not every unit conversion makes this possible. No angular unit could evenly divide both degrees and radians, for example. In these instances, there is no uniquely defined notion of a “common unit”.

In our vector space representation, we can easily compute the magnitude of the common unit by taking the smallest exponent, across all participating magnitudes, for each individual basis vector — as long as we remember to use the implicit “0” exponent for any basis vector that is omitted.

The following example may help make this clear. If we use unit as notation to represent “the common unit of unit ”, and we show only the magnitudes for simplicity, here are the steps we would follow to find the magnitude of the common unit.

—

—

This procedure produces the unambiguous correct answer whenever it is well defined. It also produces an answer for irrational “ratios”, where there is no uniquely defined result. This provides the practical benefit of making it easy to compare, say, an angle in degrees to one in radians, as long as at least one of them is represented in a floating point type.

§ 6.3 Faster than lightspeed constants

In most libraries, physical constants are implemented as constant (possibly `constexpr`) quantity values. Such an approach has some disadvantages, often resulting in longer compilation times and a loss of precision.

§ 6.3.1 Simplifying constants in an equation

When dealing with equations involving physical constants, they often occur more than once in an expression. Such a constant may appear both in a numerator and denominator of a quantity equation. As we know from fundamental physics, we can simplify such an

expression by simply striking a constant out of the equation. Supporting such behavior allows a faster runtime performance and often a better precision of the resulting value.

§ 6.3.2 Physical constants as units

The [mp-units] library allows and encourages implementing physical constants as regular units. With that, the constant's value is handled at compile-time, and under favorable circumstances, it can be simplified in the same way as all other repeated units do. If it is not simplified, the value is stored in a type, and the expensive multiplication or division operations can be delayed in time until a user selects a specific unit to represent/print the data.

Such a feature often also allows the use of simpler or faster representation types in the equation. For example, instead of always multiplying a small integral value with a big floating-point constant number, we can just use the integral type all the way. Only in case a constant will not simplify in the equation, and the user will require a specific unit, such a multiplication will be lazily invoked, and the representation type will need to be expanded to facilitate that. With that, addition, subtractions, multiplications, and divisions will always be the fastest - compiled away or done on the fast arithmetic types or in out-of-order execution.

To benefit from all of the above, in the [mp-units] library, constants (defined by SI or otherwise) are implemented as units in the following way:

```
namespace si {  
  
namespace si2019 {  
  
inline constexpr struct speed_of_light_in_vacuum :  
    named_unit<"c", mag<299'792'458> * metre / second> {}  
    speed_of_light_in_vacuum;  
  
} // namespace si2019  
  
inline constexpr struct magnetic_constant :  
    named_unit<basic_symbol_text{"μ₀", "u_0"}, mag<4> * mag_pi *  
    mag_power<10, -7> * henry / metre> {} magnetic_constant;  
  
} // namespace mp_units::si
```

§ 6.3.3 Usage examples

With the above definitions, we can calculate vacuum permittivity as:

```
constexpr auto permeability_of_vacuum = 1. * si::magnetic_constant;  
constexpr auto speed_of_light_in_vacuum = 1 *  
    si::si2019::speed_of_light_in_vacuum;  
  
QuantityOf<isq::permittivity_of_vacuum> auto q = 1 /  
    (permeability_of_vacuum * pow<2>(speed_of_light_in_vacuum));
```

```
std::cout << "permittivity of vacuum = " << q << " = " << q.in(F / m) <<
"\n";
```

The above first prints the following:

```
permittivity of vacuum = 1  $\mu_0^{-1} \text{ c}^{-2}$  = 8.85419e-12 F/m
```

As we can clearly see, all the calculations above were just about multiplying and dividing the number 1 with the rest of the information provided as a compile-time type. Only when a user wants a specific SI unit as a result, the unit ratios are lazily resolved.

Another similar example can be an equation for total energy:

```
QuantityOf<isq::mechanical_energy>                                auto
    total_energy(QuantityOf<isq::momentum> auto p,
                  QuantityOf<isq::mass>
    auto m,
                  QuantityOf<isq::speed> auto c)
{
    return isq::mechanical_energy(sqrt(pow<2>(p * c) + pow<2>(m * pow<2>
(c)))));
}
```

```
constexpr auto GeV = si::giga<si::electronvolt>;
constexpr      QuantityOf<isq::speed>      auto      c      =      1.      *
    si::si2019::speed_of_light_in_vacuum;
constexpr auto c2 = pow<2>(c);
```

```
const auto p1 = isq::momentum(4. * GeV / c);
const QuantityOf<isq::mass> auto m1 = 3. * GeV / c2;
const auto E = total_energy(p1, m1, c);
```

```
std::cout << "in `GeV` and `c`:\n"
    << "p = " << p1 << "\n"
    << "m = " << m1 << "\n"
    << "E = " << E << "\n";
```

```
const auto p2 = p1.in(GeV / (m / s));
const auto m2 = m1.in(GeV / pow<2>(m / s));
const auto E2 = total_energy(p2, m2, c).in(GeV);
```

```
std::cout << "\nin `GeV`:\n"
    << "p = " << p2 << "\n"
    << "m = " << m2 << "\n"
    << "E = " << E2 << "\n";
```

```
const auto p3 = p1.in(kg * m / s);
const auto m3 = m1.in(kg);
const auto E3 = total_energy(p3, m3, c).in(J);
```

```
std::cout << "\nin SI base units:\n"
    << "p = " << p3 << "\n"
    << "m = " << m3 << "\n"
    << "E = " << E3 << "\n";
```

The above prints the following:

```
in `GeV` and `c`:  
p = 4 GeV/c  
m = 3 GeV/c2  
E = 5 GeV  
  
in `GeV`:  
p = 1.33426e-08 GeV s/m  
m = 3.33795e-17 GeV s2/m2  
E = 5 GeV  
  
in SI base units:  
p = 2.13771e-18 kg m/s  
m = 5.34799e-27 kg  
E = 8.01088e-10 J
```

§ 6.4 Equivalence

Units, dimensions, and quantity types can be checked for equivalence with operator==. However, what it means to be an equivalent entity means something different for each case here.

§ 6.4.1 Dimensions

Equivalence is the simplest to reason about in the case of dimensions. The only thing to account for here is the point when a user would like to derive its own strong type from a library-provided one.

Please note that the library never provides strong types for derived dimensions besides the `dimension_one`. For example, ISQ defines length (L) and time (T) dimensions, but there is no such thing as a speed dimension. It does not have its own symbol as well. There is only a derived dimension of speed described with LT^{-1} . This is the reason why a user should also not derive strong types from the derived dimensions.

The equality operator for dimensions can be implemented as:

```
template<Dimension Lhs, Dimension Rhs>  
[[nodiscard]] constexpr bool operator==(Lhs lhs, Rhs rhs)  
{  
    return is_same_v<Lhs, Rhs> ||  
           detail::derived_from_the_same_base_dimension(lhs, rhs);  
}
```

§ 6.4.2 Quantity types

Equality for quantity types is similar to dimensions. Again, users are allowed to derive their own types, but only from the named strong types provided by the library.

§ 6.4.3 Units

Equality for units is a bit more complicated. Not only watt (W) should be equivalent to J/s or kg m²/s³ but also litre (l) should be equivalent to cubic decimetre (dm³). This is why in this case we do not compare user-provided types but first convert each unit to its canonical representation and then we compare if the reference unit and the magnitude is the same:

```
[[nodiscard]] constexpr bool operator==(Unit auto lhs, Unit auto rhs)
{
    auto canonical_lhs = detail::get_canonical_unit(lhs);
    auto canonical_rhs = detail::get_canonical_unit(rhs);

    detail::have_same_canonical_reference_unit(canonical_lhs.reference_unit,
    canonical_rhs.reference_unit) &&
    canonical_lhs.mag == canonical_rhs.mag;
    return
}
```

§ 6.5 Ordering

Please note that the ordering for dimensions and quantity types has no physical sense.

We could entertain adding ordering for units, but this would work only for quantities having the same reference unit, which would be inconsistent with how equivalence works.

Let's see the following example:

```
constexpr Unit auto my_unit = si::second;
if constexpr (my_unit == si::metre) {
    // ...
}
if constexpr (my_unit > si::metre) {
    // ...
}
if constexpr (my_unit > si::nano(si::second)) {
    // ...
}
```

In the above code, the first check could be useful for some use cases. However, the second one is impossible to implement and should not compile. The third one could be considered useful, but the current version of [\[mp-units\]](#) does not expose such an interface to limit potential confusion. Also, it is really hard to mathematically prove that unit magnitude representation that we use in the library is greater or smaller than the other one in some cases.

§ 7 Quantities

The quantity class template is the workhorse of the library. It can be considered a generalization of `std::chrono::duration`, but is not directly compatible with it.

§ 7.1 Quantity references

Note: We know that probably the term “reference” will not survive too long in the Committee, but we couldn’t find a better name for it in the [mp-units] library (<https://github.com/mpusz/mp-units/issues/486>).

[ISO/IEC Guide 99] says:

quantity - property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and a reference. ... A reference can be a measurement unit, a measurement procedure, a reference material, or a combination of such.

In the [mp-units] library a quantity reference represents all the domain-specific meta-data about the quantity besides its representation type and its value. A Reference concept is satisfied by either of:

- an associated unit (e.g. `si::metre`),
- an instantiation of the `reference<QuantitySpec, Unit>` class template explicitly specifying the quantity type and its unit.

A reference type is implicitly created as a result of the following expression:

```
using namespace mp_units::si::unit_symbols;
constexpr auto ref = isq::height[m];
```

The above example defines a variable of type `reference<isq::height, si::metre>`.

The reference class template also exposes an arithmetic interface similar to the one that we have already discussed in case of units and quantity types. It simply forwards the operation to its quantity type and unit members. For example:

```
template<QuantitySpec auto Q, Unit auto U>
struct reference {
    template<auto Q2, auto U2>
    [[nodiscard]] friend constexpr bool operator==(reference, reference<Q2,
        U2>)
    {
        return Q == Q2 && U == U2;
    }

    template<AssociatedUnit U2>
    [[nodiscard]] friend constexpr bool operator==(reference, U2 u2)
    {
        return Q == get_quantity_spec(u2) && U == u2;
    }

    template<auto Q2, auto U2>
    [[nodiscard]] friend constexpr reference<Q * Q2, U * U2> operator*
        (reference, reference<Q2, U2>)
    {
        return {};
    }
}
```



```

template<AssociatedUnit U2>
[[nodiscard]] friend constexpr reference<Q * get_quantity_spec(U2{}), U
    * U2{}> operator*(reference, U2)
{
    return {};
}

// ...
};

```

Please note that all of the operators work on two references, or one reference and a `AssociatedUnit`.

§ 7.2 quantity class template

Based on the ISO definition above, the quantity class template has the following signature:

```

template<Reference                                     auto R,
        RepresentationOf<get_quantity_spec(R).character> Rep = double>
class quantity;

```

It has only one data member of `Rep` type. Unfortunately, this data member is publicly exposed to satisfy the C++ language requirements for structural types. Hopefully, the language rules for structural types will improve with time before this library gets standardized.

§ 7.3 Quantity construction helpers

As we already noticed in many examples above, a numerical value multiplied or divided by a `Reference` creates the value of quantity class template with the representation type and reference deduced from the types used in the expression.

We have a few options to choose from:

— simple quantities (quantities of a quantity kind)

```

quantity<si::metre / si::second, int> q1 = 42 * m / s;
quantity q2 = 42 * m / s;

static_assert(std::is_same_v<decltype(q1), decltype(q2)>);
static_assert(q1.quantity_spec == kind_of<isq::length / isq::time>);

```

— typed quantities (quantities of a specific type within a kind)

```

quantity<isq::speed[si::metre / si::second], int> q3 = 42 * m / s;
quantity q4 = 42 * isq::speed[m / s];
quantity q5 = isq::speed(42 * m / s);

static_assert(std::is_same_v<decltype(q3), decltype(q4)>);
static_assert(std::is_same_v<decltype(q3), decltype(q5)>);
static_assert(q3.quantity_spec == isq::speed);

```

In case someone doesn't like the multiply syntax or there is an ambiguity between the operator* provided by this and other libraries, a quantity can also be created with a dedicated factory function:

```
quantity q = make_quantity<isq::speed[m / s]>(42);
```

§ 7.4 Quantity constructors and value conversions

The quantity class template has a converting constructor that participates in the overload resolution only when:

- The source quantity specification is implicitly convertible to the destination one,
- both units share the same reference unit, and
- the resulting value conversion will be value-preserving.

Additionally, this constructor becomes explicit if the source representation type is not convertible to the destination one.

§ 7.4.1 Value-preserving conversions

As of today, the [mp-units] library follows [the rules of std::chrono::duration for value-preserving conversions](#). We realize that some time has passed now and maybe we can improve in this domain. However, to our knowledge, as of today, we do not have any tools we could use in the C++ Standard to improve that.

Below we describe the current approach.

```
auto q1 = 5 * km;  
std::cout << q1.in(m) << '\n';  
quantity<si::metre, int> q2 = q1;
```

The second line above converts the input quantity to be expressed in metres and prints its value. The third line converts the quantity expressed in kilometers into one measured in metres.

In both cases, it is assumed that we can convert a quantity into another one with a unit of a higher resolution. There is no protection against overflow of the representation type built into the library.

If we try similar, but this time opposite, operations to the above, both conversions should fail to compile:

```
auto q1 = 5 * m;  
std::cout << q1.in(km) << '\n'; // Compile-time error  
quantity<si::kilo<si::metre>, int> q2 = q1; // Compile-time error
```

We can't preserve the value of a source quantity when we convert it to one with a unit of a lower resolution while dealing with an integral representation type for a quantity. In the

example above, converting 5 meters would result in 0 kilometers if internal conversion is performed using regular integer arithmetic.

While this could be a valid behavior, the problem arises when the user expects to be able to convert the quantity back to the original unit without loss of information. So the library should prevent such conversions from happening implicitly; [\[mp-units\]](#) offers the named cast `value_cast` for these conversions marked as unsafe.

To make the above conversions compile, we could use a floating-point representation type:

```
auto q1 = 5. * m;    // source quantity uses `double` as a representation
                        type
std::cout << q1.in(km) << '\n';
quantity<si::kilo<si::metre>> q2 = q1;
```

or:

```
auto q1 = 5 * m;      // source quantity uses `int` as a representation
                        type
std::cout << value_cast<double>(q1).in(km) << '\n';
quantity<si::kilo<si::metre>> q2 = q1; // double by default
```

The [\[mp-units\]](#) library follows `std::chrono::duration` logic and treats floating-point types as value-preserving.

§ 7.4.2 Value-truncating conversions

Another possibility would be to force such a truncating conversion explicitly from the code:

```
auto q1 = 5 * m;      // source quantity uses `int` as a representation
                        type
std::cout << q1.force_in(km) << '\n';
quantity<si::kilo<si::metre>, int> q2 = value_cast<km>(q1);
```

The code above makes it clear that “something bad” may happen here if we are not extra careful.

Another case for truncation happens when we assign a quantity with a floating-point representation type to the one using an integral representation type for its value:

```
auto q1 = 2.5 * m;
quantity<si::metre, int> q2 = q1;
```

Such an operation should fail to compile as well. Again, to force such a truncation, we have to be explicit in the code:

```
auto q1 = 2.5 * m;
quantity<si::metre, int> q2 = value_cast<int>(q1);
```

As we can see, it is essential not to allow such truncating conversions to happen implicitly and a good physical quantities and units library should fail at compile-time in case an user

makes such a mistake.

§ 7.5 Character of a quantity

This chapter is just a short preview to the feature that will get its own paper in the future if the Committee is interested in exploring this path.

§ 7.5.1 Scalars, vectors, and tensors

[ISO/IEC 80000] explicitly states:

Scalars, vectors and tensors are mathematical objects that can be used to denote certain physical quantities and their values. They are as such independent of the particular choice of a coordinate system, whereas each scalar component of a vector or a tensor and each component vector and component tensor depend on that choice.

Such distinction is important because each quantity character represents different properties and allows different operations to be done on its quantities.

For example, imagine a physical units library that allows the creation of a speed quantity from both `length / time` and `length * time`. It wouldn't be too safe to use such a product, right?

Now we have to realize that both of the above operations (multiplication and division) are not even mathematically defined for linear algebra types such as vectors or tensors. On the other hand, two vectors can be passed as arguments to dot and cross-product operations. The result of the first one is a scalar. The second one results in a vector that is perpendicular to both input vectors. Again, it wouldn't be safe to allow replacing those two operations with each other or expect the same results from both cases. This simply can't work.

§ 7.5.2 ISQ defines quantities of all characters

While defining quantities, [ISO/IEC 80000] explicitly specifies when a quantity has a vector or tensor character. Here are some examples:

Quantity	Character	Quantity Equation
duration	scalar	<i>{base quantity}</i>
mass	scalar	<i>{base quantity}</i>
length	scalar	<i>{base quantity}</i>
path_length	scalar	<i>{base quantity}</i>
radius	scalar	<i>{base quantity}</i>
position_vector	vector	<i>{base quantity}</i>
velocity	vector	<code>position_vector / duration</code>
acceleration	vector	<code>velocity / duration</code>

Quantity	Character	Quantity Equation
force	vector	mass * acceleration
power	scalar	force · velocity
moment_of_force	vector	position_vector × force
torque	scalar	moment_of_force · {unit-vector}
surface_tension	scalar	force / length
angular_displacement	scalar	path_length / radius
angular_velocity	vector	angular_displacement / duration * {unit-vector}
momentum	vector	mass * velocity
angular_momentum	vector	position_vector × momentum
moment_of_inertia	tensor	angular_momentum ⊗ angular_velocity

In the above equations:

- $a * b$ - regular multiplication where one of the arguments has to be scalar
- a / b - regular division where the divisor has to be scalar
- $a \cdot b$ - dot product of two vectors
- $a \times b$ - cross product of two vectors
- $|a|$ - magnitude of a vector
- {unit-vector} - a special vector with the magnitude of 1
- $a \otimes b$ - tensor product of two vectors or tensors

As of now, all of the C++ physical units libraries on the market besides [mp-units] do not support the operations mentioned above. They expose only multiplication and division operators, which do not work for linear algebra-based representation types. If a user of those libraries would like to create the quantities provided in the above table properly, this would result in a compile-time error stating that multiplication and division of two linear algebra vectors is impossible.

Outside of C++, only [Pint] provides great support in this domain.

§ 7.5.3 Characters don't apply to dimensions and units

[ISO/IEC 80000] explicitly states that dimensions are orthogonal to quantity characters:

In deriving the dimension of a quantity, no account is taken of its scalar, vector, or tensor character.

Also, it explicitly states that:

All units are scalars.

§ 7.5.4 Defining vector and tensor quantities

To specify a quantity as having a vector or tensor character, a value of the `quantity_character` enumeration can be appended to the `quantity_spec` that describes it:

```
inline constexpr struct position_vector : quantity_spec<length,
    quantity_character::vector> {} position_vector;
inline constexpr struct displacement : quantity_spec<length,
    quantity_character::vector> {} displacement;
```

With the above, all the quantities derived from `position_vector` or `displacement` will have a correct character determined according to the kind of operations included in the quantity equation defining a derived quantity.

For example, velocity in the below definition will be inferred to be a vector quantity (no explicit character override is needed):

```
inline constexpr struct velocity : quantity_spec<speed, position_vector /
    duration> {} velocity;
```

§ 7.5.5 Representation types for vector and tensor quantities

As we specified before, the quantity class template is defined as follows:

```
template<Reference auto R,
    RepresentationOf<get_quantity_spec(R).character> Rep = double>
class quantity;
```

The second template parameter is constrained with a `RepresentationOf` concept that checks if the provided representation type satisfies the requirements for the character associated with this quantity type.

Unfortunately, the current version of the C++ Standard Library does not provide any types that could be used as a representation type for vector and tensor quantities. This is why users are on their own here.

However, thanks to the provided customization points, any linear algebra library type can be used as a vector or tensor quantity representation type.

To enable the usage of a user-defined type as a representation type for vector or tensor quantities, users need to provide a partial specialization of the `is_vector` or `is_tensor` customization points.

For example, here is how it can be done for the types in [P1385](#):

```
#include <matrix>

using la_vector = STD_LA::fixed_size_column_vector<double, 3>;

template<>
inline constexpr bool mp_units::is_vector<la_vector> = true;
```

With the above, we can use `la_vector` as a representation type for our quantity:

```
Quantity auto q = la_vector{1, 2, 3} * isq::velocity[m / s];
```

Please, note that the following does not work (but see the next section):

```
Quantity auto q1 = la_vector{1, 2, 3} * m / s;  
Quantity auto q2 = isq::velocity(la_vector{1, 2, 3} * m / s);  
quantity<isq::velocity[m/s]> q3{la_vector{1, 2, 3} * m / s};
```

In all the cases above, the SI unit `m / s` has an associated scalar quantity of `isq::length / isq::time`. `la_vector` is not a correct representation type for a scalar quantity so the construction fails.

§ 7.5.6 Hacking the character

Sometimes, we want to use a vector quantity, but we don't care about its direction. For example, the standard gravity acceleration constant always points down, so we might not care about this in a particular scenario. In such a case, we may want to “hack” the library to allow scalar types to be used as a representation type for scalar quantities.

For example, we can do the following:

```
template<class T>  
    requires mp_units::is_scalar<T>  
inline constexpr bool mp_units::is_vector<T> = true;
```

which says that every type that can be used as a scalar representation is also allowed for vector quantities.

Doing the above is actually not such a big “hack” as the [ISO/IEC 80000] explicitly allows it:

A vector is a tensor of the first order and a scalar is a tensor of order zero.

Despite it being allowed by [ISO/IEC 80000], for type-safety reasons, we do not allow such a behavior by default, and a user has to opt into such scenarios explicitly.

§ 7.6 Dimensionless quantities

The quantities we discussed so far always had some specific type and physical dimension. However, this is not always the case. While performing various computations, we sometimes end up with so-called “dimensionless” quantities, which [ISO/IEC Guide 99] correctly defines as quantities of dimension one:

- Quantity for which all the exponents of the factors corresponding to the base quantities in its quantity dimension are zero.
- The measurement units and values of quantities of dimension one are numbers, but such quantities convey more information than a number.

- Some quantities of dimension one are defined as the ratios of two quantities of the same kind.
- Numbers of entities are quantities of dimension one.

§ 7.6.1 Dividing two quantities of the same kind

Dividing two quantities of the same kind always results in a quantity of dimension one. However, depending on what type of quantities we divide or what their units are, we may end up with slightly different types.

In the [mp-units] library, dividing two quantities of the same dimension always results in a quantity with the dimension being `dimension_one`. This is often different for other physical units libraries, which may return a raw representation type for such cases. A raw value is also always returned from the division of two `std::chrono::duration` values.

In the initial design of this library, the resulting type of division of two quantities was their common representation type (just like `std::chrono::duration`):

```
static_assert(std::is_same_v<decltype(10 * km / (5 * km)), int>);
```

The reasoning behind it was not providing a false impression of a strong quantity type for something that looks and feels like a regular number. Also, all of the mathematic and trigonometric functions were working fine out of the box with such representation types, so we did not have to rewrite `sin()`, `cos()`, `exp()`, and others.

However, the feedback we got from the production usage was that such an approach is really bad for generic programming. It is hard to handle the result of the two quantities' division (or multiplication) as it might be either a quantity or a fundamental type. If we want to raise such a result to some power, we must use `units::pow` or `std::pow` depending on the resulting type (`units::pow` takes the power as template arguments). Those are only a few issues related to such an approach.

Moreover, suppose we divide quantities of the same dimension, but with units of significantly different magnitudes. In that case, we may end up with a really small or a huge floating-point value, which may result in losing lots of precision. Returning a dimensionless quantity from such cases allows us to benefit from all the properties of scaled units and is consistent with the rest of the library.

§ 7.6.1.1 Dividing quantities of the same type

First, let's analyze what happens if we divide two quantities of the same type:

```
constexpr QuantityOf<dimensionless> auto q = isq::height(200 * m) /
    isq::height(50 * m);
```

In such a case, we end up with a dimensionless quantity that has the following properties:

```
static_assert(q.quantity_spec == dimensionless);
static_assert(q.dimension == dimension_one);
```



```
static_assert(q.unit == one);
```

In case we would like to print its value, we would see a raw value of 4 in the output with no unit being printed.

§ 7.6.1.2 Dividing quantities of different types

Now let's see what happens if we divide quantities of the same dimension and unit but which have different quantity types:

```
constexpr QuantityOf<dimensionless> auto q = isq::work(200 * J) /  
    isq::heat(50 * J);
```

Again we end up with `dimension_one` and `one`, but this time:

```
static_assert(q.quantity_spec == isq::work / isq::heat);
```

As shown above, the result is not of a dimensionless type anymore. Instead, we get a quantity type derived from the performed quantity equation. According to the [ISO/IEC 80000], work divided by heat is the recipe for the thermodynamic efficiency quantity, thus:

```
static_assert(implicitly_convertible(q.quantity_spec,  
    isq::efficiency_thermodynamics));
```

Please note that the quantity of `isq::efficiency_thermodynamics` is of a kind dimensionless, so it is implicitly convertible to dimensionless and satisfies the `QuantityOf<dimensionless>` concept.

§ 7.6.1.3 Dividing quantities of different units

Now, let's see what happens when we divide two quantities of the same type but different units:

```
constexpr QuantityOf<dimensionless> auto q = isq::height(4 * km) /  
    isq::height(2 * m);
```

This time we still get a quantity of dimensionless type with a `dimension_one` as its dimension. However, the resulting unit is not `one` anymore:

```
static_assert(q.unit == mag_power<10, 3> * one);
```

In case we would print the text output of this quantity, we would not see a raw value of 2000, but 2 km/m.

First, it may look surprising, but this is actually consistent with the division of quantities of different dimensions. For example, if we divide $4 * \text{km} / (2 * \text{s})$, we do not expect km to be “expanded” to m before the division, right? We would expect the result of $2 * (\text{km} / \text{s})$, which is exactly what we get when we divide quantities of the same kind.

This is a compelling feature that allows us to express huge or tiny ratios without the need for big and expensive representation types. With this, we can easily define things like a

Hubble's constant that uses a unit that is proportional to the ratio of kilometers per megaparsecs, which are both units of length:

```
inline constexpr struct hubble_constant :
    named_unit<basic_symbol_text{"H0", "H_0"}, mag<ratio{701, 10}> *
    si::kilo<si::metre> / si::second / si::mega<parsec>> {
} hubble_constant;
```

§ 7.6.2 Counts of things

Another important use case for dimensionless quantities is to provide strong types for counts of things. For example:

- [ISO/IEC 80000] (part 3) provides a rotation quantity defined as the number of revolutions,
- [ISO/IEC 80000] (part 6) provides a number_of_turns_in_a_winding quantity,
- [ISO/IEC 80000] (part 13) provides a Hamming_distance quantity defined as the number of digit positions in which the corresponding digits of two words of the same length are different.

Thanks to assigning strong names to such quantities, they can be used in the quantity equation of other quantities. For example, rotational_frequency is defined by rotation / duration.

§ 7.6.3 Lack of convertibility from fundamental types

As stated [before](#), the division of two quantities of the same kind results in a quantity of dimension one. Even though it does not have a specific physical dimension it still uses units with various ratios such as one, percent, radian, degree, etc. It is essential to be explicit about which unit we want to use for such a quantity.

However, in some cases, this might look like an overkill. The [P2980R0] provides the motivation for standardizing this library and some examples to allow the reader to understand how the code looks and feels when such a library is used. [One of the examples provides the following line:](#)

```
static_assert(10 * km / (5 * km) == 2 * one);
```

Another example could be subtracting a value 1 from the dimensionless quantity:

```
const QuantityOf<isq::time> auto fill_time_left = (height / fill_level - 1
    * one) * fill_time;
```

Some physical quantities and units libraries (e.g. [Boost.Units]) provide [implicit conversions from the values of representation types to such quantities](#).

With such support, the above examples would look in the following way:

```
static_assert(10 * km / (5 * km) == 2);
```

```
const QuantityOf<isq::time> auto fill_time_left = (height / fill_level -
1) * fill_time;
```

Such simplification might look tempting, and the [mp-units] initially provided a special support that allowed the above to compile. However, in the V2 version of the library, it was removed. There are a few reasons for that:

- Such support has sense only for quantities of dimension one with a unit one. In the case of all the other units, the specific unit should still be provided.
- If we provide implicit conversions from the representation type to a quantity of dimension one with a unit one and we start depending on such a feature, we might end up with compile-time errors after refactoring the unit in a type of such a quantity. For example:

Before	After
<pre>struct my_data { quantity<dimensionless[one]> value; }; my_data data1{.value = 42}; my_data data2{.value = 42 * one};</pre>	<pre>struct my_data { quantity<dimensionless[percent]> value; }; my_data data1{.value = 42}; // Compile-time error my_data data2{.value = 42 * one}; // OK</pre>

- Such support is purely additive (it can be added later) and requires a bunch of additional overloads not only for a constructor but for arithmetic operators and comparisons as well.

Please also note that we will still need to use explicit units to create a quantity for some cases. For example, in the following code, `asin(-1)` would use the overload from the `<math>` header of the C++ standard library rather than the one for quantities provided in `<mp-units/math.h>` header file:

```
REQUIRE_THAT(asin(-1 * one), AlmostEquals(-90. * deg));
```

§ 7.6.4 Predefined units of the dimensionless quantity

As we observed above, the most common unit for dimensionless quantities is one. It has the ratio of 1 and does not output any textual symbol.

A unit one is special in the entire type system of units as it is considered to be an identity operand in the unit expression templates. This means that, for example:

```
static_assert(one * one == one);
static_assert(one * si::metre == si::metre);
static_assert(si::metre / si::metre == one);
```

The same is also true for `dimension_one` and `dimensionless` in the domains of dimensions and quantity specifications, respectively.

Besides the unit one, there are a few other scaled units predefined in the library for usage with dimensionless quantities:

```
inline constexpr struct percent : named_unit<"%", mag<ratio{1, 100}> *  
    one> {} percent;  
inline constexpr struct per_mille : named_unit<basic_symbol_text{"‰",  
    "%o"}, mag<ratio(1, 1000)> * one> {} per_mille;
```

§ 7.6.5 Angular quantities

Special, often controversial, examples of dimensionless quantities are the angular measure and solid angular measure quantities that are defined in [ISO/IEC 80000] (part 3) to be the result of a division of `arc_length / radius` and `area / pow<2>(radius)` respectively. Moreover, [ISO/IEC 80000] also explicitly states that both can be expressed in the unit one. This means that both `isq::angular_measure` and `isq::solid_angular_measure` should be of a kind of dimensionless.

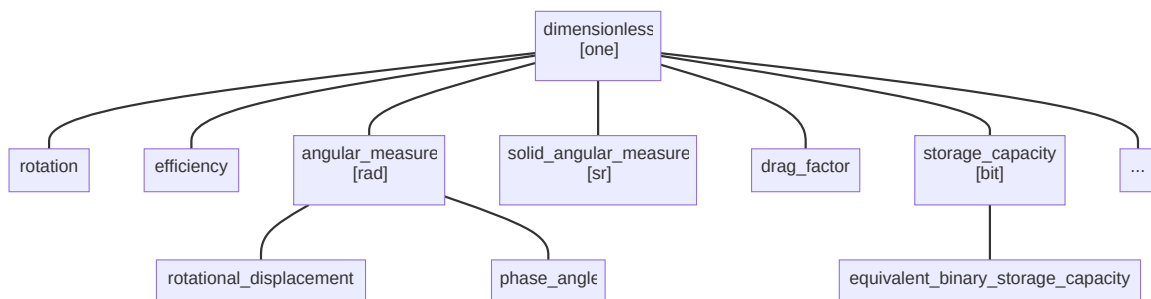
On the other hand, [ISO/IEC 80000] also specifies that the unit radian can be used for `isq::angular_measure`, and the unit steradian can be used for `isq::solid_angular_measure`. Those should not be mixed or used to express other types of dimensionless quantities. This means that both `isq::angular_measure` and `isq::solid_angular_measure` should also be quantity kinds by themselves.

Many people claim that angle being a dimensionless quantity is a bad idea. There are proposals submitted to make an angle a base quantity and rad to become a base unit in both [SI] and [ISO/IEC 80000].

§ 7.6.6 Nested quantity kinds

Angular quantities are not the only ones with such a “strange” behavior. A similar case is the `storage_capacity` quantity specified in [ISO/IEC 80000] (part 13) that again allows expressing it in both one and bit units.

Those cases make dimensionless quantities an exceptional tree in the library. This is the only quantity hierarchy that contains more than one quantity kind in its tree:



To provide such support in the library, we provided an `is_kind` specifier that can be appended to the quantity specification:

```

inline constexpr struct angular_measure : quantity_spec<dimensionless,
    arc_length / radius, is_kind> {} angular_measure;
inline constexpr struct solid_angular_measure :
    quantity_spec<dimensionless, area / pow<2>(radius), is_kind> {}
    solid_angular_measure;
inline constexpr struct storage_capacity : quantity_spec<dimensionless,
    is_kind> {} storage_capacity;

```

With the above, we can constrain radian, steradian, and bit to be allowed for usage with specific quantity kinds only:

```

inline constexpr struct radian : named_unit<"rad", metre / metre,
    kind_of<isq::angular_measure>> {} radian;
inline constexpr struct steradian : named_unit<"sr", square(metre) /
    square(metre), kind_of<isq::solid_angular_measure>> {} steradian;
inline constexpr struct bit : named_unit<"bit", one,
    kind_of<storage_capacity>> {} bit;

```

This still allows the usage of one (possibly scaled) for such quantities.

§ 7.7 Quantity arithmetics

§ 7.7.1 quantity is a numeric wrapper

If we think about it, the quantity class template is just a “smart” numeric wrapper. It exposes properly constrained set of arithmetic operations on one or two operands.

Every single arithmetic operator is exposed by the quantity class template only if the underlying representation type provides it as well and its implementation has proper semantics (e.g. returns a reasonable type).

For example, in the following code, `-a` will compile only if `MyInt` exposes such an operation as well:

```

quantity a = MyInt{42} * m;
quantity b = -a;

```

Assuming that:

- `q` is our quantity,
- `qq` is a quantity implicitly convertible to `q`,
- `q2` is any other quantity,
- `kind` is a quantity of the same kind as `q`,
- `one` is a quantity of `dimension_one` with the unit one,
- `number` is a value of a type “compatible” with `q`’s representation type,

here is the list of all the supported operators:

- unary:

- $+q$
- $-q$
- $++q$
- $q++$
- $--q$
- $q--$
- compound assignment:
 - $q += qq$
 - $q -= qq$
 - $q \% = qq$
 - $q *= \text{number}$
 - $q *= \text{one}$
 - $q /= \text{number}$
 - $q /= \text{one}$
- binary:
 - $q + \text{kind}$
 - $q - \text{kind}$
 - $q \% \text{kind}$
 - $q * q2$
 - $q * \text{number}$
 - $\text{number} * q$
 - $q / q2$
 - q / number
 - number / q
- ordering and comparison:
 - $q == \text{kind}$
 - $q <=> \text{kind}$

As we can see, there are plenty of operations one can do on a value of a quantity type. As most of them are obvious, in the following chapters, we will discuss only the most important or non-trivial aspects of quantity arithmetics.

§ 7.7.2 Addition and subtraction

Quantities can easily be added or subtracted from each other:

```
static_assert(1 * m + 1 * m == 2 * m);
static_assert(2 * m - 1 * m == 1 * m);
static_assert(isq::height(1 * m) + isq::height(1 * m) == isq::height(2 * m));
static_assert(isq::height(2 * m) - isq::height(1 * m) == isq::height(1 * m));
```

The above uses the same types for LHS, RHS, and the result, but in general, we can add, subtract, or compare the values of any quantity type as long as both quantities are of the same kind. The result of addition and subtraction will be the common type of the arguments:

```
static_assert(1 * km + 1.5 * m == 1001.5 * m);
static_assert(isq::height(1 * m) + isq::width(1 * m) == isq::length(2 * m));
static_assert(isq::height(2 * m) - isq::distance(0.5 * m) == 1.5 * m);
static_assert(isq::radius(1 * m) - 0.5 * m == isq::radius(0.5 * m));
```

Please note that for the compound assignment operators, both arguments have to either be of the same type or the RHS has to be implicitly convertible to the LHS, as the type of LHS is always the result of such an operation:

```
static_assert((1 * m += 1 * km) == 1001 * m);
static_assert((isq::height(1.5 * m) -= 1 * m) == isq::height(0.5 * m));
```

If we break those rules, the code will not compile:

```
static_assert((1 * m -= 0.5 * m) == 0.5 * m); // Compile-time error (1)
static_assert((1 * km += 1 * m) == 1001 * m); // Compile-time error (2)
static_assert((isq::height(1 * m) += isq::length(1 * m)) == 2 * m); // Compile-time error (3)
```

1. Floating-point to integral representation type is considered narrowing.
2. Conversion of quantity with integral representation type from a unit of a higher resolution to the one with a lower resolution is considered narrowing.
3. Conversion from a more generic quantity type to a more specific one is considered unsafe.

§ 7.7.3 Multiplication and division

Multiplying or dividing a quantity by a number does not change its quantity type or unit. However, its representation type may change. For example:

```
static_assert(isq::height(3 * m) * 0.5 == isq::height(1.5 * m));
```

Unless we use a compound assignment operator, in which case truncating operations are again not allowed:

```
static_assert((isq::height(3 * m) *= 0.5) == isq::height(1.5 * m)); // Compile-time error (1)
```

1. Floating-point to integral representation type is considered narrowing.

However, suppose we multiply or divide quantities of the same or different types, or we divide a raw number by a quantity. In that case, we most probably will end up in a quantity of yet another type:

```
static_assert(120 * km / (2 * h) == 60 * km / h);
static_assert(isq::width(2 * m) * isq::length(2 * m) == isq::area(4 * m2));
static_assert(50 / isq::time(1 * s) == isq::frequency(50 * Hz));
```

An exception from the above rule happens when one of the arguments is a dimensionless quantity. If we multiply or divide by such a quantity, the quantity type will not change. If such a quantity has a unit one, also the unit of a quantity will not change:

```
static_assert(120 * m / (2 * one) == 60 * m);
```

An interesting special case happens when we divide the same quantity kinds or multiply a quantity by its inverted type. In such a case, we end up with a dimensionless quantity.

```
static_assert(isq::height(4 * m) / isq::width(2 * m) == 2 * one); // (1)!
static_assert(5 * h / (120 * min) == 0 * one); // (2)!
static_assert(5. * h / (120 * min) == 2.5 * one);
```

1. The resulting quantity type of the LHS is `isq::height / isq::width`, which is a quantity of the dimensionless kind.
2. The resulting quantity of the LHS is `0 * dimensionless[h / min]`. To be consistent with the division of different quantity types, we do not convert quantity values to a common unit before the division.

§ 7.7.3.1 Beware of integral division

The physical units library can't do any runtime branching logic for the division operator. All logic has to be done at compile-time when the actual values are not known, and the quantity types can't change at runtime.

If we expect `120 * km / (2 * h)` to return `60 km/h`, we have to agree with the fact that `5 * km / (24 * h)` returns `0 km/h`. We can't do a range check at runtime to dynamically adjust scales and types based on the values of provided function arguments.

This is why we often prefer floating-point representation types when dealing with units. Some popular physical units libraries even [forbid integer division at all](#).

§ 7.7.4 Modulo

Now that we know how addition, subtraction, multiplication, and division work, it is time to talk about modulo. What would we expect to be returned from the following quantity equation?

```
auto q = 5 * h % (120 * min);
```


Most of us would probably expect to see 1 h or 60 min as a result. And this is where the problems start.

The C++ language defines its / and % operators with the [quotient-remainder theorem](#):

```
q = a / b;  
r = a % b;  
q * b + r == a;
```

The important property of the modulo operator is that it only works for integral representation types (it is undefined what modulo for floating-point types means). However, as we saw in the previous chapter, integral types are tricky because they often truncate the value.

From the quotient-remainder theorem, the result of modulo operation is $r = a - q * b$. Let's see what we get from such a quantity equation on integral representation types:

```
const quantity a = 5 * h;  
const quantity b = 120 * min;  
const quantity q = a / b;  
const quantity r = a - q * b;  
  
std::cout << "remainder: " << r << "\n";
```

The above code outputs:

```
remainder: 5 h
```

And now, a tough question needs an answer. Do we really want modulo operator on physical units to be consistent with the quotient-remainder theorem and return 5 h for $5 * h \% (120 * min)$?

This is exactly why we decided not to follow this hugely surprising path in the [\[mp-units\]](#) library. The selected approach was also consistent with the feedback from C++ experts. For example, this is what Richard Smith said about this issue:

I think the quotient-remainder property is a less important motivation here than other factors – the constraints on % and / are quite different, so they lack the inherent connection they have for integers. In particular, I would expect that A / B works for all quantities A and B, whereas $A \% B$ is only meaningful when A and B have the same dimension. It seems like a nice-to-have for the property to apply in the case where both / and % are defined, but internal consistency of / across all cases seems much more important to me.

I would expect $61 \text{ min} \% 1 \text{ h}$ to be 1 min, and $1 \text{ h} \% 59 \text{ min}$ to also be 1 min, so my intuition tells me that the result type of $A \% B$, where A and B have the same dimension, should have the smaller unit of A and B (and if the smaller one doesn't divide the larger one, we should either use the `gcd / std::common_type` of the units of A and B or perhaps just produce an error). I think any other behavior for % is hard to defend.

On the other hand, for division it seems to me that the choice of unit should probably not affect the result, and so if we want that $5 \text{ mm} / 120 \text{ min} = 0 \text{ mm/min}$, then $5 \text{ h} / 120 \text{ min} == 0 \text{ hc}$ (where hc is a dimensionless “hexaconta”, or 60x, unit). I don’t like the idea of taking SI base units into account; that seems arbitrary and like it would do the wrong thing as often as it does the right thing, especially when the units have a multiplier that is very large or small. We could special-case the situation of a dimensionless quantity, but that could lead to problematic overflow pretty easily: a calculation such as $10 \text{ s} * 5 \text{ GHz} * 2 \text{ uW}$ would overflow an int if it produces a dimensionless quantity for $10 \text{ s} * 5 \text{ GHz}$, but it could equally produce $50 \text{ G} * 2 \text{ uW} = 100 \text{ kW}$ without any overflow, and presumably would if the terms were merely reordered.

If people want to use integer-valued quantities, I think it’s fundamental that you need to know what the units of the result of an operation will be, and take that into account in how you express computations; the simplest rule for heterogeneous operators like `*` or `/` seems to be that the units of the result are determined by applying the operator to the units of the operands – and for homogeneous operators like `+` or `%`, it seems like the only reasonable option is that you get the `std::common_type` of the units of the operands.

To summarize, the modulo operator on physical units has more in common with addition and division operators than with the quotient-remainder theorem. To avoid surprising results, the operation uses a common unit to do the calculation and provide its result:

```
static_assert(5 * h / (120 * min) == 0 * one);
static_assert(5 * h % (120 * min) == 60 * min);
static_assert(61 * min % (1 * h) == 1 * min);
static_assert(1 * h % (59 * min) == 1 * min);
```

§ 7.7.5 Comparison against zero

Zero is special. It is the only number that unambiguously defines the value of any kind of quantity, regardless of its units: zero inches and zero meters and zero miles are all identical. For this reason, it’s very common to compare the value of a quantity against zero — for example, when checking the sign of a quantity, or when making sure that it’s nonzero.

We could implement such checks in the following way:

```
if (q1 / q2 != 0 * m / s)
    // ...
```

The above would work (assuming we are dealing with the quantity of speed), but it’s not ideal. If the result of `q1 / q2` is not expressed in `m / s`, we’ll incur an extra unit conversion. Even if it is in `m / s`, it’s cumbersome to repeat the unit in a context where it makes no difference.

We could avoid repeating the unit, and guarantee there won't be an extra conversion, by writing:

```
if (auto q = q1 / q2; q != q.zero())  
    // ...
```

But that is a bit inconvenient, and inexperienced users could be unaware of this technique and its reasons.

§ 7.7.5.1 Named comparison functions

For the above reasons, the [\[mp-units\]](#) library provides dedicated interfaces to compare against zero that follow the naming convention of [named comparison functions](#) in the C++ Standard Library:

- `is_eq_zero`
- `is_neq_zero`
- `is_lt_zero`
- `is_gt_zero`
- `is_lteq_zero`
- `is_gteq_zero`

Thanks to them, to save typing and not pay for unneeded conversions, our check could be implemented as follows:

```
if (is_neq_zero(q1 / q2))  
    // ...
```

Those functions will work with any type τ that exposes a `zero()` member function returning something comparable to τ . Thanks to that, we can use them not only with quantities but also with `std::chrono::duration` or any other type that exposes such an interface.

This approach has a downside, though: it produces a set of new APIs which users must learn. Nor are these six the only such functions that will need to exist: for example, `max` and `min` are perfectly reasonable to use with $\emptyset$ regardless of the units, but supporting them under this strategy would require adding a new utility function for each — and coming up with a name for those functions.

It also introduces small opportunities for error and diffs that are harder to review, because we're replacing a pattern that uses an operator (say, `a > 0`) with a named function call (say, `is_gt_zero(a)`).

These pitfalls motivate us to consider other approaches as well.

§ 7.7.5.2 zero type

The [Au] library takes a different approach to this problem. It provides an empty type, `Zero`, which represents a value of exactly 0 (in any units). It also provides an instance `ZERO` of this type. Every quantity is implicitly constructible from `Zero`.

Consider this example legacy (i.e., pre-units-library) code:

```
if (speed_squared_m2ps2 > 0) { /* ... */ }
```

When users upgrade to a units library, they will replace the raw number `speed_squared_m2ps2` with a strongly typed quantity `speed_squared`. Unfortunately, this replacement won't compile, because quantities can't be constructed from raw numeric values such as 0. They can fix this problem by using the instance, `ZERO`, which encodes its value *in the type*:

```
if (speed_squared > ZERO) { /* ... */ }
```

This has significant advantages. It preserves the *form* of the code, making the transition less error prone than replacement with a function such as `is_gt_zero`. It also reduces the number of new comparison APIs a user must learn: `Zero` handles them all.

`zero` has one downside: it will not work when passed across *generic quantity* interfaces. `Zero`'s value comes in situations where the surrounding context makes it unambiguous which quantity type it should construct. While it converts to any *specific* quantity type, it is not itself a quantity. This could confuse users.

This downside manifests in several different ways. Here are some examples:

1. While refactoring the [mp-units] code to try out this approach, we found out a perfectly reasonable place where we could not replace the numerical value 0 with `ZERO`:

```
mssl_altitude alt = mean_sea_level + 0 * si::metre; // OK
mssl_altitude alt = mean_sea_level + ZERO;           // Compile-time
error
```

This would not work because the `mean_sea_level` is an absolute point origin that stores the information about the quantity type but not its value or unit, which `mssl_altitude` needs, but none of which `ZERO` has.

2. Callsites passing `ZERO` can add friction when refactoring a concrete interface to be more generic.

```
namespace v1 { void foo(quantity<si::metre> q); }
namespace v2 { void foo(QuantityOf<isq::length> auto q); }

v1::foo(ZERO); // OK
v2::foo(ZERO); // Compile-time error
```

In practice, this issue will be discovered at the point of refactoring, so it mainly affects library authors, not their clients. They can handle this by adding an overload

for Zero, if appropriate. However, this wouldn't scale well for APIs with *multiple* parameters where users would want to pass Zero.

3. For completeness, we mention that Zero works for addition but not multiplication. When multiplying, we do not know what units (or even what dimension!) is desired for the result. However, this is not a problem in practice because users would not be motivated to write this in the first place, as simple multiplication with 0 (including any necessary units, if the result has a different dimension) would work.

```
quantity q1 = 1 * m / s;  
quantity q2 = q1 + 0 * m / s;    // OK  
quantity q3 = q1 * (0 * s);      // OK
```

```
quantity q1 = 1 * m / s;  
quantity q2 = q1 + ZERO;          // OK  
quantity q3 = q1 * ZERO;          // Compile-time error
```

The main concern with the Zero feature is that novices might be tempted to replace every numeric value 0 with the instance ZERO, becoming confused when it doesn't work. We could address this with easy-to-read documentation that clarifies its use cases and mental models.

§ 7.7.5.3 Summary (comparison against zero)

Overall, these two approaches — special functions, and a Zero type — represent two local optima in design space. Each has its strengths and weaknesses; each makes different tradeoffs. It's currently an open question as to which approach would be best suited for a quantity type in the standard library.

§ 7.7.6 Other maths

This chapter scoped only on the quantity type's operators. However, there are many named math functions provided in the [mp-units] library. Among others, we can find there the following:

- pow(), sqrt(), and cbrt(),
- exp(),
- abs(),
- epsilon(),
- floor(), ceil(), round(),
- inverse(),
- hypot(),
- sin(), cos(), tan(),
- asin(), acos(), atan().

In the library, we can also find the `<mp-units/random.h>` header file with all the pseudo-random number generators.

We plan to provide a separate paper on those in the future.

§ 8 Generic Interfaces

Using a concrete unit in the interface often has a lot of sense. It is especially useful if we store the data internally in the object. In such a case, we have to select a specific unit anyway.

For example, let's consider a simple storage tank:

```
class StorageTank {
    quantity<horizontal_area[m2]> base_;
    quantity<isq::height[m]> height_;
    quantity<isq::mass_density[kg / m3]> density_ = air_density;
public:
    constexpr StorageTank(const quantity<horizontal_area[m2]>& base, const
        quantity<isq::height[m]>& height) :
        base_(base), height_(height)
    {
    }

    // ...
};
```

As the quantities provided in the function's interface are then stored in the class, there is probably no sense in using generic interfaces here.

§ 8.1 The issues with unit-specific interfaces

However, in many cases, using a specific unit in the interface is counterproductive. Let's consider the following function:

```
quantity<isq::speed[km / h]> avg_speed(quantity<isq::length[km]> distance,
                                       quantity<isq::time[h]> duration)
{
    return distance / duration;
}
```

Everything seems fine for now. It also works great if we call it with:

```
quantity<isq::speed[km / h]> s1 = avg_speed(220 * km, 2 * h);
```

However, if the user starts doing the following:

```
quantity<isq::speed[mi / h]> s2 = avg_speed(140 * mi, 2 * h);
quantity<isq::speed[m / s]> s3 = avg_speed(20 * m, 2 * s);
```

some issues start to be clearly visible:

1. The arguments must be converted to units mandated by the function's parameters at each call. This involves potentially expensive multiplication/division operations at runtime.
2. After the function returns the speed in a unit of km/h, another potentially expensive multiplication/division operations have to be performed to convert the resulting quantity into a unit being the derived unit of the initial function's arguments.
3. Besides the obvious runtime cost, some unit conversions may result in a data truncation, which means that the result will not be exactly equal to a direct division of the function's arguments.
4. We have to use a floating-point representation type (the `quantity` class template by default uses `double` as a representation type) which is considered value preserving. Trying to use an integral type in this scenario will work only for `s1`, while `s2` and `s3` will fail to compile. Failing to compile is a good thing here as the library tries to prevent the user from doing a clearly wrong thing. To make the code compile, the user needs to use dedicated `value_cast` or `force_in` like this:

```
quantity<isq::speed[mi / h]> s2 = avg_speed(value_cast<km>(140 * mi),
      2 * h);
quantity<isq::speed[m / s]> s3 = avg_speed((20 * m).force_in(km), (2
      * s).force_in(h));
```

But the above will obviously provide an incorrect behavior (e.g. division by 0 in the evaluation of `s3`).

§ 8.2 A naive solution

A naive solution here would be to implement the function as an unconstrained function template:

```
auto avg_speed(auto distance, auto duration)
{
    return distance / duration;
}
```

Beware that there are better solutions than this. The above code is too generic. Such a function template accepts everything:

- quantities of other types
 - the compiler will not prevent accidental reordering of the function's arguments
 - quantities of different types can be passed as well
- plain `double` arguments
- `std::vector` and `std::lock_guard` will be accepted as well (of course, this will fail in the function's body later in the compilation process)

§ 8.3 Constraining function parameters with concepts

Much better generic code can be implemented using basic concepts provided with the library:

```
auto avg_speed(QuantityOf<isq::length> auto distance,
               QuantityOf<isq::time> auto duration)
{
    return isq::speed(distance / duration);
}
```

This explicitly states that the arguments passed by the user must not only satisfy a Quantity concept, but also that their quantity specification must be implicitly convertible to `isq::length` and `isq::time`, respectively. This no longer leaves room for error while still allowing the compiler to generate the most efficient code.

Please, note that now it is safe just to use integral types all the way, which again improves the runtime performance as the multiplication/division operations are often faster on integral rather than floating-point types.

§ 8.4 Constraining the function return type

The above function template resolves all of the issues described before. However, we can do even better here by additionally constraining the return type:

```
QuantityOf<isq::speed> auto avg_speed(QuantityOf<isq::length> auto
                                     distance,
                                     QuantityOf<isq::time> auto duration)
{
    return isq::speed(distance / duration);
}
```

Doing so has two important benefits:

1. It informs the users of our interface about what to expect to be the result of a function invocation. It is superior to just returning `auto`, which does not provide any hint about the thing being returned there.
2. Such a concept constrains the type returned from the function. This means that it works as a unit test to verify if our function actually performs what it is supposed to do. If there is an error in the quantity equation, we will learn about it right away.

§ 8.5 Constraining a variable on the stack

If we know exactly what the function does in its internals and if we know the exact argument types passed to such a function, we often know the exact type that will be returned from its invocation.

However, if we care about performance, we should often use the generic interfaces described in this chapter. A side effect is that we sometimes are unsure about the return type. Even if we know it today, it might change a week from now due to some code refactoring.

In such cases, we can again use `auto` to denote the type:

```
auto s1 = avg_speed(220 * km, 2 * h);  
auto s2 = avg_speed(140 * mi, 2 * h);  
auto s3 = avg_speed(20 * m, 2 * s);
```

In this case, it is probably OK to do so as the `avg_speed` function name explicitly provides the information on what to expect as a result.

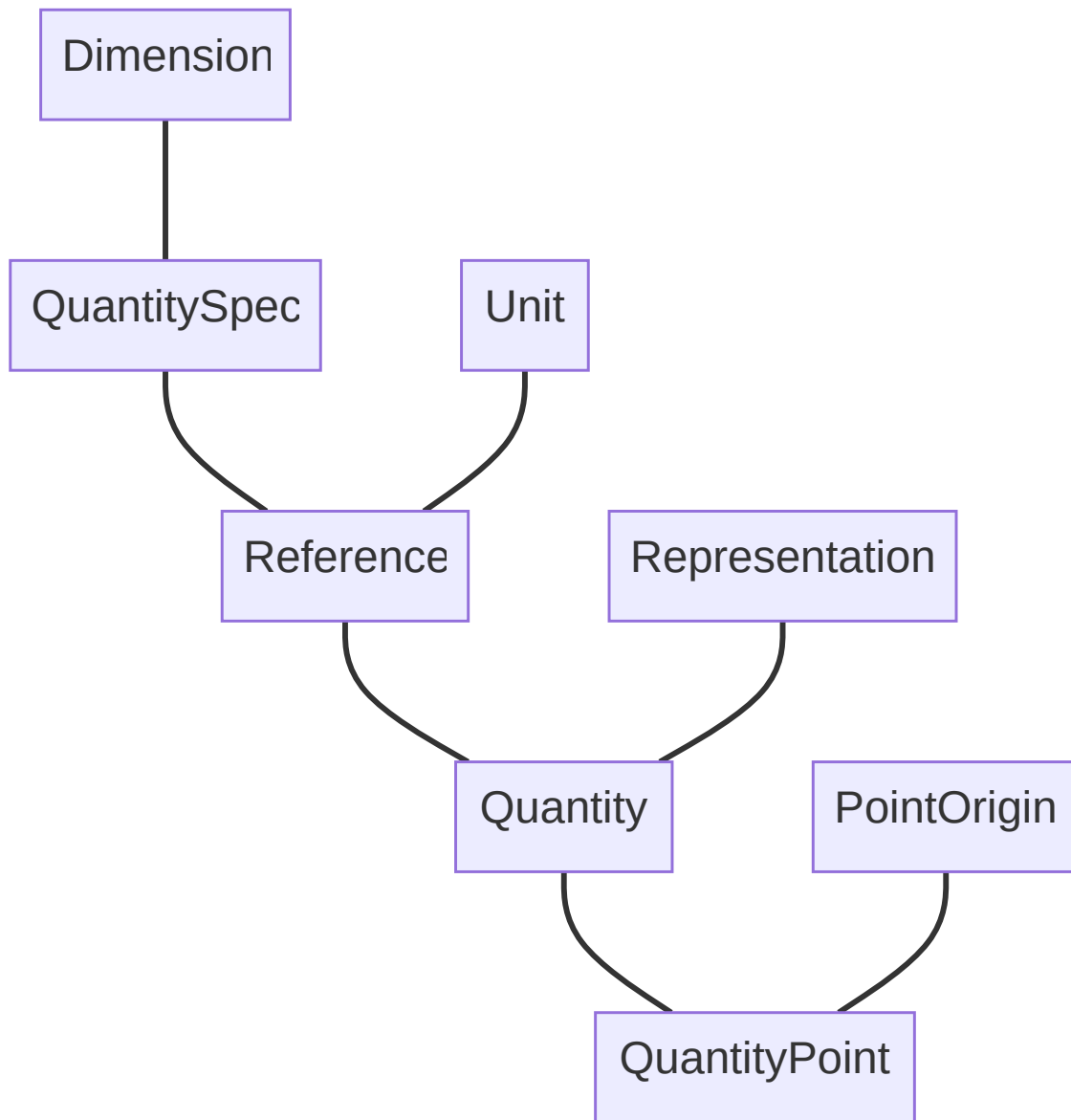
In other scenarios where the returned quantity type is not so obvious, it is again helpful to constrain the type with a concept like so:

```
QuantityOf<isq::speed> auto s1 = avg_speed(220 * km, 2 * h);  
QuantityOf<isq::speed> auto s2 = avg_speed(140 * mi, 2 * h);  
QuantityOf<isq::speed> auto s3 = avg_speed(20 * m, 2 * s);
```

Again, this explicitly provides additional information about the quantity we are dealing with in the code, and it serves as a unit test checking if the “thing” returned from a function is actually what we expected here.

§ 8.6 Basic concepts

Here are some more concepts exposed in the library and dependencies between them:



§ 9 Custom representation types

The library of physical quantities and units library should work with any custom representation type. Those can be used to:

- Improve safety (e.g., prevent overflows, restrict the range of accepted values, etc.),
- provide additional information (e.g., not only a quantity value but also the uncertainty of the measurement), and
- enable linear algebra usage.

As of right now, we have two other concurrent proposals to SG6 in this subject on the fly ([P2993] and [P3003R0]), so we do not provide any concrete requirements or recommendations here. Based on the results of discussions on the mentioned proposals, we will provide correct guidelines in the next revisions of this paper.

§ 10 The affine space

The affine space has two types of entities:

- **point** - a position specified with coordinate values (e.g. location, address, etc.)
- **vector** - the difference between two points (e.g. shift, offset, displacement, duration, etc.)

The *vector* described here is specific to the affine space theory and is not the same thing as the quantity of a vector character that we discussed in the before (although, in some cases, those terms may overlap).

§ 10.1 Operations in the affine space

Here are the primary operations one can do in the affine space:

- $vector + vector \rightarrow vector$
- $vector - vector \rightarrow vector$
- $-vector \rightarrow vector$
- $vector * scalar \rightarrow vector$
- $scalar * vector \rightarrow vector$
- $vector / scalar \rightarrow vector$
- $point - point \rightarrow vector$
- $point + vector \rightarrow point$
- $vector + point \rightarrow point$
- $point - vector \rightarrow point$

It is not possible to:

- add two *points*,
- subtract a *point* from a *vector*,
- multiply nor divide *points* with anything else.

§ 10.2 Vector is modeled by quantity

Up until now, each time when we used a quantity in our code, we were modeling some kind of a difference between two things:

- the distance between two points
- duration between two time points
- the difference in speed (even if relative to 0)

As we already know, a quantity type provides all operations required for the *vector* type in an affine space.

§ 10.3 *Point* is modeled by `PointOrigin` and `quantity_point`

In the `[mp-units]` library, the *point* abstraction is modeled by:

- The `PointOrigin` concept that specifies a measurement's origin, and
- the `quantity_point` class template that specifies a *point* relative to a specified origin.

§ 10.3.1 Absolute *point* origin

The **absolute point origin** specifies where the “zero” of our measurement's scale is. Users can specify such an origin by deriving from the `absolute_point_origin` class template:

```
constexpr struct mean_sea_level : absolute_point_origin<isq::altitude> {}  
    mean_sea_level;
```

§ 10.3.2 `quantity_point`

The `quantity_point` class template specifies an absolute quantity with respect to an origin:

```
template<Reference auto R,  
        PointOriginFor<get_quantity_spec(R)> auto P0,  
        RepresentationOf<get_quantity_spec(R).character> Rep = double>  
class quantity_point;
```

As we can see above, the `quantity_point` class template exposes one additional parameter compared to `quantity`. The `P0` parameter satisfies a `PointOriginFor` concept and specifies the origin of our measurement scale.

As a *point* can be represented with a *vector* from the origin, a `quantity_point` class template can be created with the following operations:

```
quantity_point qp1 = mean_sea_level + 42 * m;  
quantity_point qp2 = 42 * m + mean_sea_level;  
quantity_point qp3 = mean_sea_level - 42 * m;
```

It is not allowed to subtract a *point* from a *vector*, thus `42 * m - mean_sea_level` is an invalid operation.

Similarly to the creation of a quantity, if someone does not like the operator-based syntax to create a `quantity_point`, the same results can be achieved with the `make_quantity_point` factory function:

```
quantity_point qp4 = make_quantity_point<mean_sea_level>(42 * m);  
quantity_point qp5 = make_quantity_point<mean_sea_level>(-42 * m);
```

The provided quantity representing an offset from the origin is stored inside the `quantity_point` class template and can be obtained with the `quantity_from_origin()` member function:

```
constexpr quantity_point everest_base_camp_alt = mean_sea_level +
    isq::altitude(5364 * m);
static_assert(everest_base_camp_alt.quantity_from_origin() == 5364 * m);
```

§ 10.3.3 Relative *point* origin

We often do not have only one ultimate “zero” point when we measure things.

Continuing the Mount Everest trip example above, measuring all daily hikes from the `mean_sea_level` might not be efficient. Maybe we know that we are not good climbers, so all our climbs can be represented with an 8-bit integer type allowing us to save memory in our database of climbs? Why not use `everest_base_camp_alt` as our reference point?

For this purpose, we can define a `relative_point_origin` in the following way:

```
constexpr struct everest_base_camp :
    relative_point_origin<everest_base_camp_alt> {} everest_base_camp;
```

The above can be used as an origin for subsequent *points*:

```
constexpr quantity_point first_climb_alt = everest_base_camp +
    isq::altitude(std::uint8_t{42} * m);
static_assert(first_climb_alt.quantity_from_origin() == 42 * m);
```

As we can see above, the `quantity_from_origin()` member function returns a relative distance from the current point origin. In case we would like to know the absolute altitude that we reached on this climb, we can subtract the absolute point origin from the current *point*:

```
static_assert(first_climb_alt - mean_sea_level == 5406 * m);
static_assert(first_climb_alt - first_climb_alt.absolute_point_origin ==
    5406 * m);
```

§ 10.3.4 Converting between different representations of the same *point*

As we might represent the same *point* with *vectors* from various origins, the **mp-units** library provides facilities to convert the *point* to the `quantity_point` class templates expressed in terms of different origins.

For this purpose, we can either use:

— A converting constructor:

```
constexpr quantity_point<isq::altitude[m], mean_sea_level, int> qp =
    first_climb_alt;
static_assert(qp.quantity_from_origin() == 5406 * m);
```

— A dedicated conversion interface:

```
constexpr quantity_point qp =
    first_climb_alt.point_for(mean_sea_level);
static_assert(qp.quantity_from_origin() == 5406 * m);
```

It is only allowed to convert between various origins defined in terms of the same `absolute_point_origin`. Even if it is possible to express the same *point* as a *vector* from another `absolute_point_origin`, the library will not provide such a conversion. A custom user-defined conversion function will be needed to add this functionality.

Said another way, in the `[mp-units]` library, there is no way to spell how two distinct `absolute_point_origin` types relate to each other.

§ 10.3.5 *Point arithmetics*

Let's assume we will attend the CppCon conference hosted in Aurora, CO, and we want to estimate the distance we will travel. We have to take a taxi to a local airport, fly to DEN airport with a stopover in FRA, and, in the end, get a cab to the Gaylord Rockies Resort & Convention Center:

```
constexpr struct home : absolute_point_origin<isq::distance> {} home;

quantity_point<isq::distance[km], home> home_airport = home + 15 * km;
quantity_point<isq::distance[km], home> fra_airport = home_airport + 829 *
    km;
quantity_point<isq::distance[km], home> den_airport = fra_airport + 8115 *
    km;
quantity_point<isq::distance[km], home> cppcon_venue = den_airport + 10.1
    * mi;
```

As we can see above, we can easily get a new point by adding a quantity to an origin or another quantity point.

If we want to find out the distance traveled between two points, we simply subtract them:

```
quantity<isq::distance[km]> total = cppcon_venue - home;
quantity<isq::distance[km]> flight = den_airport - home_airport;
```

If we would like to find out the total distance traveled by taxi as well, we have to do a bit more calculations:

```
quantity<isq::distance[km]> taxi1 = home_airport - home;
quantity<isq::distance[km]> taxi2 = cppcon_venue - den_airport;
quantity<isq::distance[km]> taxi = taxi1 + taxi2;
```

Now, if we print the results:

```
std::cout << "Total distance: " << total << "\n";
std::cout << "Flight distance: " << flight << "\n";
std::cout << "Taxi distance: " << taxi << "\n";
```

we will see the following output:

Total distance: 8975.25 km
Flight distance: 8944 km
Taxi distance: 31.2544 km

It is not allowed to subtract two point origins defined in terms of `absolute_point_origin` (e.g. `mean_sea_level - mean_sea_level`). Those do not contain information about the unit, so we are not able to determine the resulting quantity type.

§ 10.3.6 Temperature support

Another important example of relative point origins is support of temperature quantity points in units different than kelvin (symbol K).

The [SI] definition in the [mp-units] library provides two predefined point origins:

```
namespace si {  
  
inline          constexpr          struct          absolute_zero          :  
    absolute_point_origin<isq::thermodynamic_temperature>          {}  
    absolute_zero;  
inline constexpr struct ice_point : relative_point_origin<absolute_zero +  
    273.15 * kelvin> {} ice_point;  
  
}
```

With the above, we can be explicit about the origin of our temperature point. For example, if we want to implement the degrees Celsius scale, we can do it as follows:

```
using Celsius_point = quantity_point<isq::Celsius_temperature[deg_C],  
    si::ice_point>;
```

Notice that while stacking point origins, we can use not only different representation types, but also different units for an origin and a *point*. In the above example, the relative point origin is defined in terms of `si::kelvin`, while the quantity point uses `si::degree_Celsius`.

To play a bit with temperatures, we can implement a simple room's AC temperature controller in the following way:

```
constexpr struct room_reference_temp : relative_point_origin<si::ice_point  
    + 21 * deg_C> {} room_reference_temp;  
using room_temp = quantity_point<isq::Celsius_temperature[deg_C],  
    room_reference_temp>;  
  
constexpr auto step_delta = isq::Celsius_temperature(0.5 * deg_C);  
constexpr int number_of_steps = 6;  
  
room_temp room_low = room_reference_temp - number_of_steps * step_delta;  
room_temp room_high = room_reference_temp + number_of_steps * step_delta;  
  
std::println("| {:<14} | {:^18} | {:^18} | {:^18} |", "Temperature", "Room  
    reference", "Ice point", "Absolute zero");  
std::println("| {0: ^16}|{0: ^20}|{0: ^20}|{0: ^20}|", "");
```

```

auto print = [&](std::string_view label, auto v){
    std::println("| {:<14} | {:^18} | {:^18} | {:^18} |",
                  label, v - room_reference_temp, v - si::ice_point, v -
                  si::absolute_zero);
};

print("Lowest", room_low);
print("Default", room_reference_temp);
print("Highest", room_high);

```

The above prints:

Temperature zero	Room reference	Ice point	Absolute
=====	=====	=====	=====
Lowest	-3 °C	18 °C	291.15 °C
Default	0 °C	21 °C	294.15 °C
Highest	3 °C	24 °C	297.15 °C

§ 10.3.7 No text output for *points*

The library does not provide text output for quantity points, as printing just a number and a unit is not enough to adequately describe a quantity point. Often, an additional postfix is required.

For example, the text output of 42 m may mean many things and can also be confused with an output of a regular quantity. On the other hand, printing 42 m AMSL for altitudes above mean sea level is a much better solution, but the library does not have enough information to print it that way by itself.

§ 10.4 The affine space is about type-safety

The following operations are not allowed in the affine space:

- **adding** two `quantity_point` objects
 - It is physically impossible to add the positions of home and Denver airport.
- **subtracting** a `quantity_point` from a quantity
 - What would it mean to subtract the DEN airport location from a distance?
- **multiplying/dividing** a `quantity_point` with a scalar
 - What is the position of 2 * DEN airport location?
- **multiplying/dividing** a `quantity_point` with a quantity
 - What would multiplying a distance with the DEN airport location mean?
- **multiplying/dividing** two `quantity_point` objects
 - What would multiplying the home and DEN airport location mean?

- **mixing** `quantity_points` of different quantity kinds
 - It is physically impossible to subtract ISQ time from ISQ length.
- **mixing** `quantity_points` of inconvertible quantities
 - What does it mean to subtract a distance point to DEN airport from the Mount Everest base camp altitude?
- **mixing** `quantity_points` of convertible quantities but with unrelated origins
 - How to subtract a point on our trip to CppCon measured relatively to our home location from a point measured relative to the center of the Solar System?

The usage of `quantity_point`, and affine space types in general, improves expressiveness and type-safety of the code we write.

§ 11 Acknowledgements

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§ 12 References

[Au] The Au Units library.

<https://aurora-opensource.github.io/au>

[Boost.Units] Matthias C. Schabel and Steven Watanabe. Boost.Units.

https://www.boost.org/doc/libs/1_83_0/doc/html/boost_units.html

[ISO/IEC 80000] ISO/IEC 80000: Quantities and units.

<https://www.iso.org/committee/46202/x/catalogue/>

[ISO/IEC Guide 99] ISO/IEC Guide 99: International vocabulary of metrology — Basic and general concepts and associated terms (VIM).

<https://www.iso.org/obp/ui#iso:std:iso-iec:guide:99>

[JCGM 200:2012] International vocabulary of metrology - Basic and general concepts and associated terms (VIM) (JCGM 200:2012, 3rd edition).

<https://jcgmbipm.org/vim/en>

[Measurement Data] William Kent, Stephanie Lechner Janowski, Bruce Hamilton, and Dan Hepner. Measurement Data (Archive Report).

<https://www.bkent.net/Doc/mdarchiv.pdf>

[mp-units] mp-units - A Physical Quantities and Units library for C++.

<https://mpusz.github.io/mp-units>

[P2980R0] Mateusz Pusz, Dominik Berner, Johel Ernesto Guerrero Peña, Charles Hogg, Nicolas Holthaus, Roth Michaels, Vincent Reverdy. 2023-10-15. A motivation, scope, and plan for a physical quantities and units library.

<https://wg21.link/p2980r0>

- [P2982R0] Mateusz Pusz, Chip Hogg. 2023-10-15. ``std::quantity`` as a numeric type.
<https://wg21.link/p2982r0>
- [P2993] Luke Valenty. Constrained Numbers.
<https://wg21.link/p2993>
- [P3003R0] Johel Ernesto Guerrero Peña. 2023-10-14. The design of a library of number concepts.
<https://wg21.link/p3003r0>
- [Pint] Pint: makes units easy.
<https://pint.readthedocs.io/en/stable/index.html>
- [SI] SI Brochure: The International System of Units (SI).
<https://www.bipm.org/en/publications/si-brochure>